Optimal Transport in linear Independent Component Analysis

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Declaration of Academic Integrity

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Chapter 1

Introduction

Blind source separation, specifically formulated as independent component analysis (ICA), is a fundamental problem in statistics (Jutten & Herault, 1985). The objective of ICA is to recover a set of hidden, statistically independent source signals from an observed set of mixtures. In this thesis, only linear mixtures, wherein the observations are assumed to be generated by an unknown, invertible linear combination of the latent sources are considered.

To achieve this separation, ICA imposes statistical independence among the recovered components. This approach is frequently contrasted with principal component analysis (PCA). While PCA diagonalizes the covariance matrix to ensure that the resulting components are uncorrelated, ICA enforces full statistical independence (Hyvärinen, Karhunen, & Oja, 2001, Chapter 1). This stronger criterion allows ICA to disentangle complex latent variables that remain mixed under simple decorrelation.

1.1 Existing Methodologies

The standard ICA pipeline begins by centering and whitening the observed mixture. Whitening is a necessary preprocessing step that removes linear correlations and scales the marginal variances to unity. Once the data is whitened, the search for independent components is restricted to finding an optimal orthogonal rotation matrix on a space of orthogonal matrices. (Absil, Mahony, & Sepulchre, 2008).

Traditional ICA algorithms navigate this rotational search by optimizing contrast functions to identify non-Gaussian projections (Hyvärinen et al., 2001, Chapter 8). These classical methods rely on surrogate information-theoretic measures, such as negentropy approximations or cumulant-based matching, to quantify the non-Gaussianity of the unmixed components (Amari, Cichocki, & Yang, 1996).

While computationally efficient, these parametric approximations evaluate specific statistical moments rather than the full probability density. Consequently, we find that they fail to capture the complete geometric structure of the underlying distributions, leading to optimization failures in heterogeneous mixtures.

1.2 Applications

The ability to blindly separate independent sources without parametric assumptions regarding their underlying distributions has widespread utility across diverse domains. In neuroscience, ICA serves as the standard clinical technique for isolating and removing high-amplitude biological artifacts, such as ocular and muscular signals, from continuous electroencephalogram (EEG) and magnetoencephalogram (MEG) recordings (Hyvärinen et al., 2001, Chapter 21). In telecommunications, ICA is deployed to separate overlapping transmissions in code-division multiple access (CDMA) systems, while in image processing, it is utilized to extract localized, oriented, and bandpass features from natural images, effectively mimicking the processing functions of the mammalian primary visual cortex (Hyvärinen et al., 2001, Chapters 22-23).

Furthermore, the methodology is increasingly utilized in advanced econometrics and market microstructure. In financial price discovery, ICA facilitates the identification of latent structural shocks, allowing researchers to uniquely resolve the orthogonal ambiguities inherent in standard vector error correction models (Zema & Cordoni, 2025). Across all these distinct domains, the validity of the downstream scientific or economic analysis depends strictly on the reliable convergence of the underlying ICA optimization algorithm.

1.3 Main Contribution

The primary methodological contribution of this thesis is the formulation and investigation of an optimal transport theory based framework for independent component analysis (OT-ICA). The core objective of the OT-ICA framework reduces to searching the space of orthogonal matrices for the rotation matrix that maximizes the non-Gaussianity of the projected data.

To achieve this, we propose utilizing the Wasserstein distance as the contrast function for the rotational search. [1] Our objective is to find the orthogonal candidates that maximize the Wasserstein distance to a standard normal distribution. Because the Wasserstein metric evaluates the global geometry of the empirical distribution using order statistics (Villani, 2003), it provides a non-parametric

[1]The complete implementation of the OT-ICA framework is publicly available at: `https://gitfront.io/r/ashutosh-jha/4gwwN1sPeeAD/ot-in-linear-ica/`

and highly sensitive measure of distance to Gaussianity. We find that this helps to circumvent the structural blind spots of existing methods which use surrogate information-theoretic or moment based contrasts.

1.4 Structure of the Thesis

The remainder of this thesis is structured as follows. Chapter 2 establishes the theoretical foundations of linear ICA, information geometry, and optimal transport. Chapter 3 formalizes the OT-ICA framework, and proves that the squared Wasserstein (W_2^2) distance reliably bounds mixture non-Gaussianity. Chapter 4 details the algorithmic enhancements required to optimize this metric efficiently on the Stiefel manifold. Chapter 5 presents our empirical evaluation, demonstrating the failure modes of standard proxy optimization and how OT-ICA addresses these limitations. Finally, Chapter 6 concludes the thesis with a summary of findings and directions for future research.

Chapter 2

Theoretical Foundations

This chapter establishes the foundations necessary for linear independent component analysis. We first formalize the linear mixing model and its identifiability conditions, subsequently detailing the statistical intuitions that motivate the framework. Following this, we introduce an information-geometric perspective linking statistical independence to non-Gaussianity, and finally, formalize the principles of optimal transport that govern the proposed optimization metric.

2.1 Linear Independent Component Analysis (ICA)

The foundational assumption of the linear Independent Component Analysis model is that the observed signals are generated by a linear combination of statistically independent latent sources. We formalize this generative model and its required constraints as follows:

Theorem 2.1 The Linear ICA Model and Identifiability, (Comon, 1994). *Let the observed data be denoted by a random vector $\mathbf{X} \in \mathbb{R}^d$, and the latent sources by $\mathbf{S} \in \mathbb{R}^d$. The linear ICA model assumes:*

*a) **Linear Mixing:** $\mathbf{X} = \mathbf{A}\mathbf{S}$, where $\mathbf{A} \in \mathbb{R}^{d \times d}$ is an unknown, invertible mixing matrix.*

b) The components $s_1, s_2, \ldots, s_d$ of $\mathbf{S}$ are mutually statistically independent.

c) At most one of the source components s_i follows a Gaussian distribution.

Under these conditions, the mixing matrix $\mathbf{A}$ can be uniquely identified from the observations $\mathbf{X}$, up to a permutation and scaling of its columns.

Expanded into vector notation, the structural mixing represents the observed variables as

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1d} \\ a_{21} & a_{22} & \cdots & a_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ a_{d1} & a_{d2} & \cdots & a_{dd} \end{bmatrix} \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_d \end{bmatrix}. \tag{2.1}$$

The core objective of ICA is to estimate an unmixing matrix $\mathbf{B} \approx \mathbf{A}^{-1}$ such that the recovered components $\mathbf{Z} = \mathbf{BX}$ are as statistically independent as possible, thereby reconstructing the original sources $\mathbf{S}$.

2.1.1 The Insufficiency of Principal Component Analysis

Principal Component Analysis (PCA) is frequently utilized as a baseline technique to decorrelate observed data. This corresponds to finding the orthogonal matrix $\mathbf{V}^\top$ derived from the eigenvalue decomposition of the data's covariance matrix. However, rendering a sample strictly decorrelated is mathematically insufficient for achieving blind source separation when the underlying data is non-Gaussian.

As noted by MacKay (2003), PCA is fundamentally an L_2-norm variance maximization technique; it is inherently sensitive to variable scaling and heavily influenced by outliers. More critically, PCA relies exclusively on second-order statistics. Once the data is decorrelated and scaled (whitened) such that $\mathbb{E}[\mathbf{XX}^\top] = \mathbf{I}_d$, the PCA criterion is satisfied for any arbitrary orthogonal rotation. While decorrelation implies strict independence for multivariate Gaussian distributions, higher-order dependencies remain intact for non-Gaussian data. Consequently, ICA must transcend second-order moments, traversing the space of orthogonal matrices to locate the rotation that eliminates these higher-order dependencies and achieves statistical independence.

2.1.2 Identifiability and Ambiguities

The non-Gaussianity constraint defined in Theorem 2.1 is a strict mathematical requirement. If two or more sources are Gaussian, any orthogonal transformation of those specific sources yields another set of strictly independent Gaussian variables, rendering the mixing matrix $\mathbf{A}$ impossible to determine uniquely.

Furthermore, even when the non-Gaussianity assumption holds, the model has two inherent ambiguities. The first is a scaling ambiguity: because both $\mathbf{A}$ and $\mathbf{S}$ are unknown, any scalar multiplier assigned to a source s_i can be perfectly cancelled by dividing the corresponding column of $\mathbf{A}$ by the same scalar. Therefore, the exact variances of the unobserved sources cannot be inferred. The second is a permutation ambiguity: the model treats the sum of sources symmetrically, meaning the indexing order of the recovered components is arbitrary.

2.1.3 Centering and Whitening

To simplify the optimization landscape, the observed data $\mathbf{X}$ is subjected to a two-step preprocessing phase consisting of centering and whitening. Centering ensures that the data has a zero mean by subtracting the expectation, $\mathbb{E}[\mathbf{X}] = \mathbf{0}$. Given centered data, the covariance matrix simplifies to $\mathrm{Cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$. Whitening transforms the observed vector into a new vector $\widetilde{\mathbf{X}}$ whose components are uncorrelated and possess unit variance. This is achieved using the eigenvalue decomposition of the covariance matrix

$$\mathrm{Cov}(\mathbf{X}) = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^\top \tag{2.2}$$

where $\mathbf{V}$ is the orthogonal matrix of eigenvectors and $\boldsymbol{\Lambda}$ is the diagonal matrix of eigenvalues λ_j. The whitening transformation matrix $\mathbf{W}$ is constructed by scaling the eigenvectors by the inverse square root of the eigenvalues

$$\mathbf{W} = \boldsymbol{\Lambda}^{-1/2}\mathbf{V}^\top. \tag{2.3}$$

Applying this transformation yields the whitened data $\widetilde{\mathbf{X}} = \mathbf{W}\mathbf{X}$, where the covariance is the identity matrix $\mathbf{I}_d$. After whitening, if we define the unmixing operation on the whitened data as $\widetilde{\mathbf{Z}} = \mathbf{U}\widetilde{\mathbf{X}}$, the covariance of the recovered components is

$$\mathrm{Cov}(\widetilde{\mathbf{Z}}) = \mathbf{U}\mathrm{Cov}(\widetilde{\mathbf{X}})\mathbf{U}^\top = \mathbf{U}\mathbf{I}_d\mathbf{U}^\top = \mathbf{U}\mathbf{U}^\top. \tag{2.4}$$

Because the independent components must have unit variance, we require $\mathbf{U}\mathbf{U}^\top = \mathbf{I}_d$. This reveals that the unmixing matrix $\mathbf{U}$ must be perfectly orthogonal, thus reducing the ICA optimization to a search over the Stiefel manifold, a space of orthogonal matrices.

2.1.4 Statistical and Thermodynamic Intuitions

The identifiability limitation in Theorem 2.1 leads to the translation of the challenge of finding independent components into a distinct optimization objective: maximizing non-Gaussianity. The theoretical justification for this approach is rooted in the Central Limit Theorem.

Theorem 2.2 (Central Limit Theorem Intuition). *Let $s_1, s_2, \ldots, s_d$ be independent random variables. Under general regularity conditions, the distribution of their normalized sum tends toward a Gaussian distribution as d increases (Billingsley, 1995).*

While Theorem 2.2 formally applies to limits of sums, its consequence in ICA is that any observed linear mixture of independent, non-Gaussian sources will inherently appear more Gaussian than the underlying sources themselves. (Hyvärinen et al., 2001, Chapter 1). Therefore, by extracting the unmixed components that are the least Gaussian, the algorithm effectively isolates the original independent signals.

The use of non-Gaussianity as a proxy for independence is further understood from information-theoretic and statistical-mechanics perspectives. For any continuous random variable with a fixed finite variance, it is a foundational mathematical result that the Gaussian distribution uniquely possesses the maximum possible Shannon entropy (Cover & Thomas, 2006, Chapter 8). In thermodynamics, isolated physical systems naturally evolve toward states of maximum entropy. Viewed through this interdisciplinary lens, the Central Limit Theorem serves as the statistical manifestation of this physical principle: the mathematical process of linearly mixing independent sources inherently increases the overall entropy of the system, inevitably driving the joint distribution toward the high-entropy Gaussian state.

2.2 An Information Geometry Perspective on ICA

2.2.1 Product and Gaussian Manifolds

Information geometry provides a framework for quantifying the relationship between independence and non-Gaussianity by analyzing probability distributions as points on geometric manifolds. Following the perspective introduced by Cardoso (2022), ICA can be understood as orthogonally projecting the empirical distribution onto two distinct subspaces. The first is the Product manifold $\mathcal{P}$, which contains all distributions where the marginal components are strictly independent. The second is the Gaussian manifold $\mathcal{G}$, which encompasses all multivariate Gaussian distributions with zero mean and a fixed covariance structure. We dive into this perspective with the knowledge that standard decorrelation merely ensures that the covariance is diagonal, but full ICA seeks to find a linear transform that minimizes the divergence between the empirical data and its projection onto the Product manifold (Cardoso, 2022).

2.2.2 Kullback-Leibler Pythagorean Identities

Distances between these distributions are measured using the Kullback-Leibler (KL) divergence (Cover & Thomas, 2006). For any random vector Y with a joint density P_Y, the mutual information $\mathscr{I}(Y)$ measures the distance to the closest

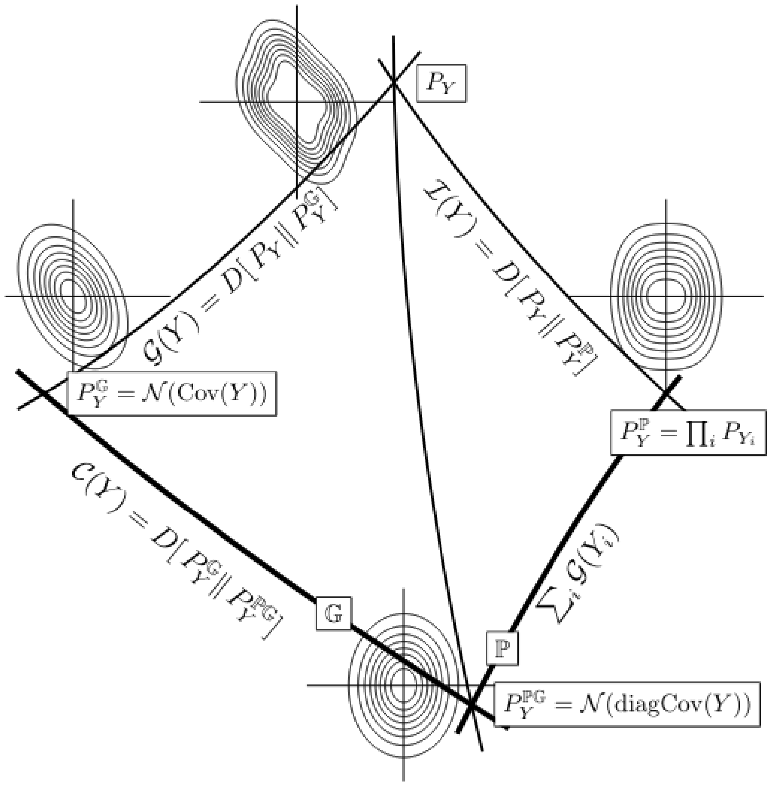

Figure 2.1: *The Pythagorean geometry of Independent Component Analysis. The empirical joint distribution P_Y is orthogonally projected onto the Product manifold $\mathcal{P}$ (yielding the independent marginals P_Y^P) and the Gaussian manifold $\mathcal{G}$ (yielding the best Gaussian fit $\mathcal{N}(\mathrm{Cov}\, Y)$). The intersection of these approximations yields the fully factorized Gaussian target $\mathcal{N}(\mathrm{diag}\,\mathrm{Cov}\, Y)$. The Kullback-Leibler divergences between these nodes geometrically represent distinct statistical properties: Mutual Information $\mathscr{I}(Y)$, Joint Non-Gaussianity $G(Y)$, and Correlation $C(Y)$. Figure from Cardoso (2022).*

independent product distribution, denoted $P_Y^P = \prod_i P_{Y_i}$. This expands integrally as:

$$\mathscr{I}(Y) = D_{\mathrm{KL}}\left(P_Y \,\|\, P_Y^P\right) = \int P_Y(y) \log \frac{P_Y(y)}{\prod_i P_{Y_i}(y_i)}\, dy. \tag{2.5}$$

A Pythagorean identity exists on the Product manifold. For any candidate product distribution $Q = \prod_i q_i$, the divergence splits precisely into the mutual information and the marginal divergences. Expanding the logarithm reveals this additive structure:

$$\begin{aligned} D_{\mathrm{KL}}\left(P_Y \| Q\right) &= \int P_Y(y) \log \left(\frac{P_Y(y)}{P_Y^P(y)} \frac{P_Y^P(y)}{Q(y)} \right) dy \\ &= \int P_Y(y) \log \frac{P_Y(y)}{P_Y^P(y)}\, dy + \sum_i \int P_{Y_i}(y_i) \log \frac{P_{Y_i}(y_i)}{q_i(y_i)}\, dy \\ &= \mathscr{I}(Y) + \sum_i D_{\mathrm{KL}}\left(P_{Y_i} \| q_i\right). \end{aligned} \tag{2.6}$$

A similar Pythagorean identity holds for the Gaussian manifold $\mathcal{G}$. Let $\mathcal{N}(\Sigma)$ denote an arbitrary zero-mean multivariate Gaussian distribution parameterized by a covariance matrix Σ. Furthermore, let $\mathcal{N}(\operatorname{Cov} Y)$ denote the specific Gaussian distribution that shares the exact covariance structure of the empirical data Y. Geometrically, $\mathcal{N}(\operatorname{Cov} Y)$ represents the orthogonal projection of P_Y onto the Gaussian manifold, yielding the best possible Gaussian approximation of the data.

The KL divergence from the empirical distribution P_Y to the arbitrary Gaussian target $\mathcal{N}(\Sigma)$ decomposes into the divergence to this best Gaussian fit and the divergence between the two Gaussian models:

$$D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\Sigma)\big) = D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\operatorname{Cov} Y)\big) + D_{\mathrm{KL}}\big(\mathcal{N}(\operatorname{Cov} Y) \| \mathcal{N}(\Sigma)\big). \tag{2.7}$$

2.2.3 Independence, Correlation, and Non-Gaussianity

These geometric identities can be combined to relate independence directly to non-Gaussianity. We define the non-Gaussianity of the joint variable Y as $G(Y) = D_{\mathrm{KL}}(P_Y \,\|\, \mathcal{N}(\operatorname{Cov} Y))$, and the scalar measure of correlation as $C(Y) = D_{\mathrm{KL}}(\mathcal{N}(\operatorname{Cov} Y) \,\|\, \mathcal{N}(\operatorname{diag} \operatorname{Cov} Y))$.

To establish the fundamental balance between these quantities, we evaluate the KL divergence from the empirical distribution P_Y to a fully factorized Gaussian target $Q = \mathcal{N}(\operatorname{diag} \operatorname{Cov} Y)$. We can compute this divergence via two distinct geometric paths shown in Figure 2.1.

Following the Pythagorean identity on the Product manifold (Equation 2.6):

$$D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\operatorname{diag} \operatorname{Cov} Y)\big) = \mathscr{I}(Y) + \sum_i G(Y_i). \tag{2.8}$$

Following the Pythagorean identity on the Gaussian manifold (Equation 2.7):

$$D_{\mathrm{KL}}\big(P_Y \| \mathcal{N}(\operatorname{diag} \operatorname{Cov} Y)\big) = G(Y) + C(Y). \tag{2.9}$$

Equating these two paths yields Cardoso's unified theorem (Cardoso, 2022):

$$\mathscr{I}(Y) + \sum_i G(Y_i) = G(Y) + C(Y). \tag{2.10}$$

To demonstrate how maximizing marginal non-Gaussianities minimizes mutual information, we analyze the terms under the constraints of the ICA pipeline. Crucially, the joint non-Gaussianity $G(Y)$ is invariant under any invertible linear transformation (Cardoso, 2022). Furthermore, once the data has been whitened, the covariance matrix is forced to be the identity matrix. This implies that $\operatorname{Cov}(Y) = \operatorname{diag} \operatorname{Cov}(Y)$, which strictly forces the correlation term to zero ($C(Y) = 0$).

Applying these constraints, the relationship rearranges into a direct subtraction:

$$\begin{aligned} \mathscr{I}(Y) + \sum_i G(Y_i) &= G(Y) + 0 \\ \mathscr{I}(Y) &= G(Y) - \sum_i G(Y_i). \end{aligned} \tag{2.11}$$

Because $G(Y)$ remains constant under orthogonal rotation, minimizing the mutual information $\mathscr{I}(Y)$, finding independent components, is mathematically equivalent to maximizing the sum of the marginal non-Gaussianities $\sum_i G(Y_i)$. This shows that centering and whitening constrain the data to a subspace where true statistical independence can be achieved solely by maximizing non-Gaussianity.

2.3 Optimal Transport Theory

2.3.1 The Monge Problem: Push-Forwards and Pull-Backs

While information geometry utilizes Kullback-Leibler divergence, optimal transport provides an alternative metric space parameterized by the physical geometry of probability mass. First formulated by Gaspard Monge (1781), the problem was originally conceptualized as finding the most cost-effective way to move a distribution of mass.

Mathematically, this is framed as finding a transport map T that associates points from a continuous source measure μ to a target measure ν. We express this spatial modification of probability mass using the *push-forward* operator $T_\#$. A measure ν is the push-forward of μ by a map T, denoted $\nu = T_\#\mu$, if for any measurable set B in the target space:

$$\nu(B) = \mu(T^{-1}(B)). \tag{2.12}$$

As illustrated in Figure 2.2, the push-forward operator acts on measures, effectively pushing mass forward along the map T. It is conceptually distinct from the *pull-back* operator $T^\#$, which acts on functions. For a continuous, bounded test function $h : \mathcal{Y} \to \mathbb{R}$, the pull-back evaluates the function via the composition mapping ($T^\# h = h \circ T$). Because push-forward and pull-back are adjoint operators, the Monge problem seeks to minimize a transportation cost function $c(x, y)$ subject to the strict constraint that $T_\#\mu = \nu$:

$$\min_T \left\{ \int_{\mathcal{X}} c(x, T(x)) d\mu(x) \quad : \quad T_\#\mu = \nu \right\}. \tag{2.13}$$

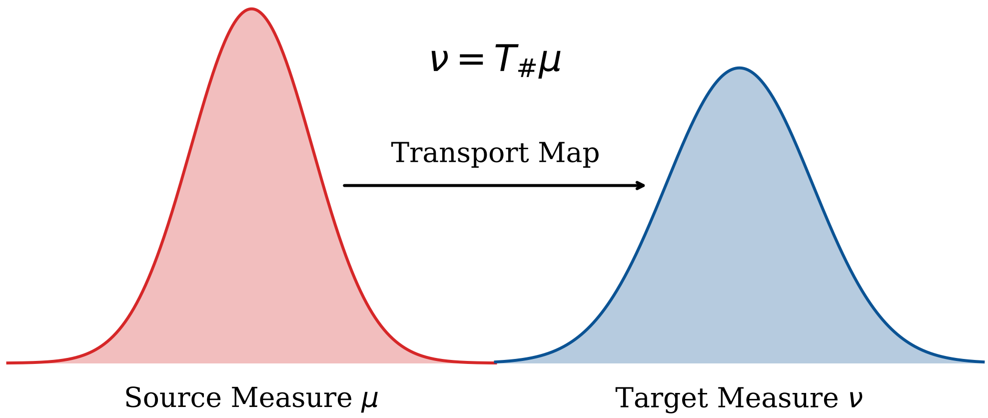

Figure 2.2: *The Monge transport map T pushes forward the probability mass from the source measure μ to construct the target measure ν, strictly conserving total mass during the transformation.*

2.3.2 The Kantorovich Relaxation

The fundamental limitation of the Monge problem is that the map T must be strictly deterministic; mass from a single source location cannot be split to multiple targets. Consequently, if the target distribution has more discrete points than the source distribution, a valid Monge map may not even exist.

To resolve this degeneracy, Kantorovich (1942) relaxed the deterministic requirement, proposing instead a probabilistic transport that permits mass splitting. The Kantorovich formulation replaces the deterministic map T with a joint probability distribution, or coupling, $\pi \in \Pi(\mu, \nu)$, which dictates exactly how fractions of mass are distributed from the source to the target.

As illustrated in Figure 2.3, this relaxation is universally applicable across three fundamental statistical scenarios. In the discrete setting depicted in the first column, the coupling manifests as a transportation matrix mapping distinct point masses. In the semi-discrete setting (second column), mass from a continuous density is optimally partitioned into specific discrete target regions. In the fully continuous setting (third column), the coupling forms a joint density over the continuous supports. By allowing fractional transport in all domains, Kantorovich transformed the non-convex combinatorial Monge problem into a relatively tractable linear optimization problem.

2.3.3 The Wasserstein Distances (W_1 and W_2)

The minimal cost required to transform one probability distribution into another is formalized as the Wasserstein distance. For continuous measures in arbitrary

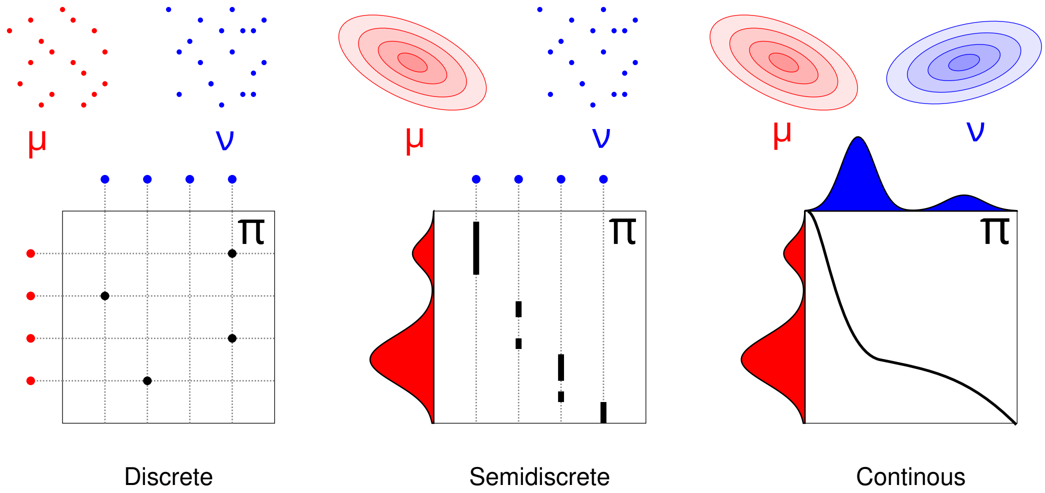

Figure 2.3: *A schematic view of the input measures and optimal couplings across the three primary scenarios of Kantorovich Optimal Transport: Discrete, Semi-discrete, and Continuous. The top row depicts the marginal distributions, while the bottom row visualizes the structural nature of the corresponding coupling spaces. Figure adapted from Peyré and Cuturi (2019).*

dimensions d, utilizing a Euclidean distance cost function $c(x, y) = \|x - y\|^p$, the general W_p distance is defined over the space of optimal couplings $\Pi(\mu, \nu)$ as:

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu,\nu)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^p \, d\pi(x, y) \right)^{1/p}. \tag{2.14}$$

While the integral definition above correctly formalizes optimal transport in high dimensions, direct analytical computation of $\pi(x, y)$ is highly restrictive for arbitrary multivariable spaces. The concept of sorting data to construct a cumulative distribution function (CDF) is strictly a one-dimensional phenomenon, as there is no natural, universal ordering of coordinates in $\mathbb{R}^d$ $(d > 1)$.

However, in the specific one-dimensional setting, the geometry of optimal transport drastically simplifies (Villani, 2003, Chapter 2). Because the data can be unambiguously sorted, the W_1 distance analytically collapses to the integral of the absolute difference between the univariate cumulative distribution functions F_μ and F_ν:

$$W_1(\mu, \nu) = \int_{-\infty}^{\infty} |F_\mu(x) - F_\nu(x)| \, dx. \tag{2.15}$$

Similarly, the squared Wasserstein distance $(p = 2)$, which heavily penalizes local geometric distortions, takes on a highly efficient analytical form in one dimension. It is most conveniently expressed using the quantile functions, which are the exact inverses of the cumulative distributions (F_μ^{-1} and F_ν^{-1}):

$$W_2^2(\mu, \nu) = \int_0^1 |F_\mu^{-1}(t) - F_\nu^{-1}(t)|^2 \, dt. \tag{2.16}$$

By circumventing the need to construct multi-dimensional couplings, the 1D W_2^2 metric provides an extremely computationally efficient mechanism for measuring structural divergence. When evaluating empirical linear projections against standard normal noise, the 1D squared Wasserstein distance serves as the primary optimization function for the framework utilized in this thesis.

Chapter 3

The Optimal Transport ICA (OT-ICA) Framework

3.1 Optimal Transport Distance as a Contrast for ICA

As established in the preceding chapter, the process of centering and whitening restricts the search for independent components to the manifold of orthogonal matrices. Furthermore, Cardoso's unified theorem (Cardoso, 2022) demonstrates that within this whitened space, the mutual information of the projections is minimized precisely when the sum of their marginal non-Gaussianities is maximized. Traditional Independent Component Analysis algorithms approach this optimization by utilizing proxy contrast functions, such as kurtosis or negentropy approximations. While computationally inexpensive, these proxies evaluate specific statistical moments rather than the full probability density.

This thesis formulates ICA directly as an optimal transport problem, utilizing the squared Wasserstein distance, W_2^2, as a direct measure of non-Gaussianity. We formalize this framework as follows:

Definition 3.1 The OT-ICA Optimization Objective. *Let* $\mathbf{X}$ *be the whitened observed mixture, and let* $\mathbf{u} \in S_n$ *be a candidate projection vector on the unit sphere. Let the empirical distribution of the projected data be denoted by* $\mathbb{P}_{\mathbf{u}^\top \mathbf{X}}$*, and let the standard normal distribution be defined as* $\Gamma \equiv \mathcal{N}(0, 1)$*. The Optimal Transport ICA (OT-ICA) objective is to find the orthogonal rotation matrix whose column vectors* $\mathbf{u}$ *maximize the distance to* Γ *(maximize non-gaussianity):*

$$\hat{\mathbf{u}} = \arg\max_{\mathbf{u} \in S_n} W_2^2\big(\mathbb{P}_{\mathbf{u}^\top \mathbf{X}}, \Gamma\big). \tag{3.1}$$

By framing the problem in this manner, we leverage the optimal transport metric's ability to evaluate geometric deviations from Gaussianity, providing a sensitive contrast function.

3.2 Theoretical Motivation: Bounding Mixture Non-Gaussianity

To mathematically justify the use of W_2^2 as an objective function for ICA, we demonstrate that maximizing this distance recovers the true independent sources. By the Central Limit Theorem, a mixture of non-Gaussian sources is more Gaussian than the constituent sources themselves. In optimal transport terms, this implies that the Wasserstein distance between a mixture and a Gaussian distribution is bounded by the distances of its independent components.

3.2.1 The W_2 Upper Bound

Theorem 3.1 **W_2 Upper Bound.** *Let $\mathbf{Z} = (Z_1, \ldots, Z_d)^\top$ be a vector of mutually independent latent sources, and let $\mathbf{w}$ be a weight vector on the unit sphere ($\sum_{i=1}^{d} w_i^2 = 1$). The squared Wasserstein distance between the linear mixture $\mathbf{w} \cdot \mathbf{Z}$ and the standard normal distribution Γ is bounded by the weighted sum of the source distances to Γ:*

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^{d} w_i^2 W_2^2(Z_i, \Gamma). \tag{3.2}$$

Proof. We construct a specific joint probability distribution, or coupling, using a common source of randomness. Let $\mathbf{N} = (N_1, \ldots, N_d)^\top$ be a vector of d independent standard normal variables, $\mathbf{N} \sim \mathcal{N}(0, \mathbf{I}_d)$, assumed to be statistically independent of the sources $\mathbf{Z}$. For each independent latent source Z_i, there exists an optimal transport map T_i that pushes forward the standard normal distribution to the source distribution, denoted as $(T_i)_\# \Gamma = Z_i$.

We construct a random variable X representing the mixture using the transport maps evaluated on the common noise vector:

$$X = \sum_{i=1}^{d} w_i T_i(N_i). \tag{3.3}$$

Because $T_i(N_i)$ follows the distribution of Z_i, X models the weighted mixture $\mathbf{w} \cdot \mathbf{Z}$. To compare this mixture to a Gaussian distribution, we construct a target variable

Y using the same underlying noise vector $\mathbf{N}$ and weights $\mathbf{w}$:

$$Y = \sum_{i=1}^{d} w_i N_i. \tag{3.4}$$

Given that the weights sum to unity and the components N_i are independent standard normals, the resulting variable Y follows the standard normal distribution, $Y \sim \Gamma$. The pair (X, Y) creates a valid coupling between the mixture distribution and the Gaussian target.

The squared Wasserstein distance is the infimum of the expected squared cost over all valid couplings. Therefore, the optimal distance is less than or equal to the cost of our constructed coupling (X, Y), yielding the inequality:

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \Gamma) \leq \mathbb{E}\left[|X - Y|^2\right]. \tag{3.5}$$

Expanding the squared difference yields:

$$\begin{aligned} |X - Y|^2 &= \left| \sum_{i=1}^{d} w_i T_i(N_i) - \sum_{i=1}^{d} w_i N_i \right|^2 \\ &= \left(\sum_{i=1}^{d} w_i \big(T_i(N_i) - N_i\big) \right)^2. \end{aligned} \tag{3.6}$$

Squaring the summation produces diagonal terms $(i = j)$ and cross terms $(i \neq j)$:

$$|X - Y|^2 = \sum_{i=1}^{d} w_i^2 \big(T_i(N_i) - N_i\big)^2 + \sum_{i \neq j} w_i w_j \big(T_i(N_i) - N_i\big)\big(T_j(N_j) - N_j\big). \tag{3.7}$$

Because the underlying noise variables N_i and N_j are independent, and the functions evaluate to a mean of zero, when we take the expectation, all cross terms vanish. We are left with the expectation of the diagonal terms:

$$\mathbb{E}\left[|X - Y|^2\right] = \sum_{i=1}^{d} w_i^2 \mathbb{E}\left[\big(T_i(N_i) - N_i\big)^2\right]. \tag{3.8}$$

Recognizing that $\mathbb{E}[(T_i(N_i) - N_i)^2]$ is the definition of the squared Wasserstein distance between the individual source Z_i and Γ, we arrive at the bound:

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \Gamma) \leq \sum_{i=1}^{d} w_i^2 W_2^2(Z_i, \Gamma). \quad \blacksquare \tag{3.9}$$

By constructing a valid joint coupling, we establish that cross-dimensional terms vanish under expectation, proving that mixture non-Gaussianity is bounded by the weighted sum of the constituent sources. This property acts as an optimal transport analogue to the Central Limit Theorem, demonstrating that linear mixing inherently reduces the overall distance to Gaussianity. Consequently, maximizing this distance forces the weight vector $\mathbf{w}$ to collapse toward a canonical axis, effectively isolating a single independent component.

3.2.2 Comonotonicity and the Proof of Strict Inequality

Optimizing ICA requires establishing that mixtures are strictly less non-Gaussian than the original sources. We determine the conditions under which the upper bound becomes a strict inequality.

Theorem 3.2 Strict W_2 Inequality for Non-Gaussian Sources. *Assuming at most one source Z_i is Gaussian, and the source distributions possess smooth and strictly positive densities, the upper bound derived in Theorem 3.1 is a strict inequality for any valid mixture where multiple weights w_i are non-zero:*

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^{d} w_i^2 W_2^2(Z_i, \Gamma). \tag{3.10}$$

Proof. In a one-dimensional setting, the W_2 distance equals the expected squared difference of a specific coupling if and only if the random variables are comonotonic. By Brenier's Theorem (Brenier, 1991), this implies there exists a strictly increasing, deterministic function mapping one variable to the other. For the constructed mixture variable X and the Gaussian target Y to be comonotonic with respect to the shared noise vector $\mathbf{N} \in \mathbb{R}^d$, their gradients must be parallel at every point in the probability space. This requires the existence of a scalar function $\lambda(\mathbf{N})$ such that:

$$\nabla X(\mathbf{N}) = \lambda(\mathbf{N}) \nabla Y(\mathbf{N}). \tag{3.11}$$

Assuming the source distributions possess smooth and strictly positive densities, Caffarelli's regularity theory (Caffarelli, 1992) guarantees that the optimal transport maps T_i are continuously differentiable. We compute the gradients of our constructed variables with respect to $\mathbf{N}$. The gradient of the Gaussian target is the weight vector:

$$Y(\mathbf{N}) = \sum_{i=1}^{d} w_i N_i \implies \nabla Y = \mathbf{w}. \tag{3.12}$$

The gradient of the source mixture leverages the derivatives of the individual transport maps:

$$X(\mathbf{N}) = \sum_{i=1}^{d} w_i T_i(N_i) \implies \nabla X = \begin{bmatrix} w_1 T_1'(N_1) \\ w_2 T_2'(N_2) \\ \vdots \\ w_d T_d'(N_d) \end{bmatrix}. \tag{3.13}$$

Substituting these gradients into the comonotonicity condition yields the system of equations:

$$\nabla X = \lambda(\mathbf{N})\mathbf{w}. \tag{3.14}$$

To determine if this condition holds, we differentiate both the Left-Hand Side (LHS) and the Right-Hand Side (RHS) with respect to $\mathbf{N}$ to compute their Hessian matrices.

On the LHS, because X separates into independent terms $w_i T_i(N_i)$, the cross-derivatives $\frac{\partial^2 X}{\partial N_i \partial N_j}$ are zero for all $i \neq j$. This yields a diagonal matrix containing the second derivatives of the transport maps. On the RHS, we differentiate the product $\lambda(\mathbf{N})\mathbf{w}$. Applying the product rule yields the outer product of $\mathbf{w}$ and $\nabla\lambda$, forming a Rank-1 matrix. Equating the matrices:

$$\underbrace{\begin{bmatrix} w_1 T_1''(N_1) & 0 & \cdots & 0 \\ 0 & w_2 T_2''(N_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & w_d T_d''(N_d) \end{bmatrix}}_{\text{Hessian of } X \text{ (Diagonal)}} = \underbrace{\begin{bmatrix} w_1 \frac{\partial\lambda}{\partial N_1} & w_1 \frac{\partial\lambda}{\partial N_2} & \cdots & w_1 \frac{\partial\lambda}{\partial N_d} \\ w_2 \frac{\partial\lambda}{\partial N_1} & w_2 \frac{\partial\lambda}{\partial N_2} & \cdots & w_2 \frac{\partial\lambda}{\partial N_d} \\ \vdots & \vdots & \ddots & \vdots \\ w_d \frac{\partial\lambda}{\partial N_1} & w_d \frac{\partial\lambda}{\partial N_2} & \cdots & w_d \frac{\partial\lambda}{\partial N_d} \end{bmatrix}}_{\text{Outer Product } \mathbf{w}(\nabla\lambda)^\top \text{ (Rank-1)}} \tag{3.15}$$

Examining any off-diagonal entry located at row i and column j $(i \neq j)$, the diagonal LHS matrix dictates that this entry must be exactly zero. The RHS Rank-1 matrix evaluates this entry as $w_i \frac{\partial\lambda}{\partial N_j}$, establishing the requirement:

$$w_i \frac{\partial\lambda}{\partial N_j} = 0 \quad \text{for all } i \neq j. \tag{3.16}$$

Because we assume a mixture where multiple component weights are non-zero $(w_i \neq 0)$, it follows that $\frac{\partial\lambda}{\partial N_j} = 0$. Since this holds across all dimensions, the gradient of the scaling function is zero $(\nabla\lambda = \mathbf{0})$. Consequently, $\lambda(\mathbf{N})$ is a global scalar constant, λ.

Equating the diagonal elements implies $T_i'(N_i) = \lambda$. Integrating this relationship indicates the optimal transport maps are linear functions of the form:

$$T_i(x) = \lambda x + c. \tag{3.17}$$

Applying a purely linear transformation to Gaussian noise N_i produces another Gaussian distribution. The identifiability assumption of ICA dictates that the original latent sources Z_i are non-Gaussian, which requires their optimal transport mappings from a Gaussian base to be non-linear. Therefore, the linear map requirement contradicts the non-Gaussianity of the sources. Because the condition for equality cannot be met, the relationship is a strict inequality.

$$W_2^2(\mathbf{w} \cdot \mathbf{Z}, \Gamma) < \sum_{i=1}^{d} w_i^2 W_2^2(Z_i, \Gamma). \quad \blacksquare \tag{3.18}$$

The practical implication of this strict inequality dictates the behavior of the objective function. The upper bound is structured as a convex combination of the marginal non-Gaussianities, subject to the constraint $\sum_{i=1}^{d} w_i^2 = 1$. When the objective function is maximized, the supremum is achieved when the entirety of the weight is assigned to the single latent source exhibiting the maximum distance to Gaussianity. Consequently, finding the global maximum forces the weight vector $\mathbf{w}$ to collapse to a canonical axis ($w_k^2 = 1$ and $w_{i \neq k}^2 = 0$), thereby isolating the most non-Gaussian independent component from the mixture.

3.2.3 Sequential Extraction via Orthogonal Deflation

Maximizing the W_2^2 objective isolates only the most non-Gaussian source, but the framework requires a mechanism to unmix the remaining $(d-1)$ latent components. This is achieved through a sequential deflation process.

Let $\mathbf{w}_1$ be the canonical weight vector recovered by the initial maximization. To extract the second component, the optimization must be restricted to the orthogonal complement of $\mathbf{w}_1$. We update the search space by constraining the subsequent weight vector $\mathbf{w}_2$ to be strictly orthogonal to the first ($\mathbf{w}_2^\top \mathbf{w}_1 = 0$).

For the k-th component, the maximization is performed under the constraint that the new weight vector $\mathbf{w}_k$ satisfies $\mathbf{w}_k^\top \mathbf{w}_j = 0$ for all previously extracted components $j < k$. By iteratively extracting the maximum of the objective function and restricting the subsequent search to the remaining orthogonal subspace, the algorithm sequentially constructs the full orthogonal unmixing matrix $\mathbf{W} \in \mathbb{R}^{d \times d}$. This guarantees that all d mutually independent sources are recovered.

3.3 Empirical Landscape: W_1 versus W_2^2

Having established the theoretical foundations of the W_2^2 metric, we evaluate its practical suitability for manifold optimization. In the context of ICA, gradient ascent algorithms must climb the loss landscape to locate the maxima of the distance metric, thereby isolating the independent sources.

While the W_1 distance (Earth Mover's Distance) is robust to outliers, its reliance on an L_1 absolute penalty introduces geometric challenges when applied to finite empirical samples. We demonstrate why the squared L_2 penalty of the W_2^2 metric is preferable for gradient-based convergence.

3.3.1 Empirical Gradients and Derivative Volatility

To observe this behavior, we generated a synthetic bivariate mixture of Student-t distributions ($\nu = 4$) and evaluated both the raw distances and their first-order numerical gradients across a continuous sweep of rotation angles $\theta \in [0, \pi]$. Because the unmixing matrix restricts the search to orthogonal vectors, the two true independent components are geometrically separated by exactly $\pi/2$ radians (90°).

The empirical results, visualized in Figure 3.1, illustrate the behaviors of the two metrics during continuous gradient ascent. In optimal transport, because the W_1 cost scales linearly with distance, the resulting optimization landscape lacks the convexity needed to smooth out finite sample noise.

The consequence of this lack of curvature is evident in the bottom gradient profile. Because W_1 relies on piecewise linear sorting mechanics without a squaring penalty, its empirical derivative fluctuates frequently as the projection angle rotates. Consequently, first-order gradient ascent solvers will encounter a jagged derivative surface, which hinders convergence to the true peak. This instability is exacerbated in higher dimensions, where the W_1 derivative surface becomes even more jagged and difficult to traverse. Furthermore, this volatility limits the utility of second-order Quasi-Newton approximations (such as L-BFGS), as a stable Hessian cannot be reliably estimated from such a jagged and difficult-to-traverse gradient.

In contrast, the squared penalty in the W_2^2 metric acts as a natural smoother, penalizing local geometric distortions. As seen in the gradient profiles, the W_2^2 derivative behaves continuously, decelerating predictably as it approaches the optimum. This retention of curvature provides a reliable signal for gradient-based solvers, allowing the OT-ICA framework to converge consistently.

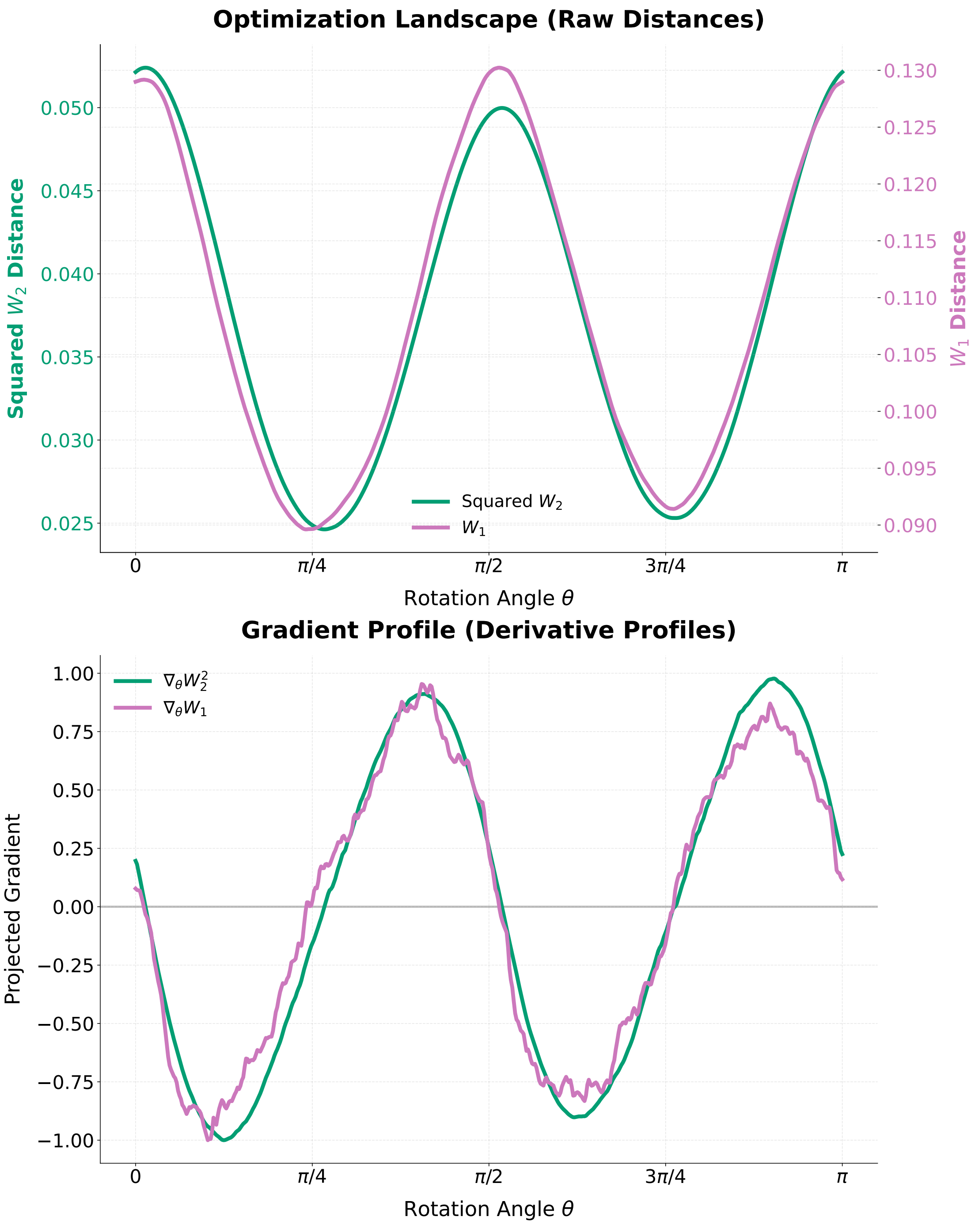

Figure 3.1: *The empirical optimization landscapes (top) and their normalized first derivatives (bottom). While both metrics peak near the true independent components ($\theta \approx 0, \pi/2, \pi$), their gradient behaviors differ. The W_2^2 gradient smoothly approaches zero, whereas the W_1 gradient exhibits jagged volatility on finite samples.*

Chapter 4

Algorithmic Enhancements

This chapter transitions the theoretical formulation of OT-ICA into a practical and efficient algorithm. While the baseline optimal transport formulation provides a valid objective, its reliance on discrete sorting and unconstrained gradient descent presents both algorithmic and computational bottlenecks. To resolve these issues, this chapter introduces a series of targeted algorithmic enhancements. We develop analytical Gaussian targets to eliminate quantile approximation noise, formalize an optimization framework on the Orthogonal Group to strictly preserve component decorrelation, and utilize various methodologies to navigate discontinuous landscapes. The chapter concludes by combining these components into the complete OT-ICA algorithm.

4.1 The Baseline OT-ICA Algorithm

Having established the theoretical validity of the squared Wasserstein distance (W_2^2) for Independent Component Analysis, the challenge is to minimize this cost function computationally. Just like the standard FastICA algorithm (Hyvärinen et al., 2001), the baseline OT-ICA algorithm proceeds by first centering and whitening the observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, yielding the whitened data $\widetilde{\mathbf{X}}$. The algorithm initializes an unmixing matrix $\mathbf{W} \in \mathbb{R}^{d \times d}$ within the Orthogonal Group, denoted hereafter as the space $\mathcal{S}$, which is the mathematical set of all orthogonal matrices.

In the forward pass, the data is projected onto the candidate components, $\mathbf{Y} = \mathbf{W}\widetilde{\mathbf{X}}$. For each row of $\mathbf{Y}$, the empirical cumulative distribution function (CDF) is constructed by sorting the N observations. This sorting operation evaluates the inverse CDF (the quantile function) over a uniform probability grid, typically approximated as $p_i = (i - 0.5)/N$ (Villani, 2003, Chapter 2). The W_2^2 distance is computed by matching these sorted empirical quantiles to the corresponding quantiles of a standard normal distribution.

While theoretically sound, the standard implementation of this gradient descent loop faces geometric limitations. The optimal transport matching requires sorting at each iteration, presenting a computational bottleneck. Furthermore, standard gradient updates do not inherently respect the strict orthogonality constraint of the mixing matrix, and discrete approximations of the target Gaussian introduce gradient noise. To make OT-ICA competitive with existing methods, specific algorithmic enhancements are required.

4.2 Computational Bottlenecks and Solutions

4.2.1 Analytical Gaussian Targets

One source of instability in the baseline algorithm is the discrete approximation of the target Gaussian distribution. Traditionally, target quantiles are evaluated at evenly spaced intervals ($q_i = \Phi^{-1}((i-0.5)/N)$). Representing a continuous Gaussian distribution with finite discrete points introduces approximation noise. During gradient computation, this noise translates into a non-smooth optimization landscape.

To eliminate this noise, we replace point-sampled targets with an analytical formulation. We compute the expected value of a standard normal variable within each discrete quantile bin.

Definition 4.1 Analytical Gaussian Target. *Let the uniform probability bin edges for N samples be defined as $p_i = \frac{i}{N}$ for $i = 0, \ldots, N$, with corresponding Gaussian domain boundaries $z_i = \Phi^{-1}(p_i)$, where Φ is the standard normal cumulative distribution function. The analytical target value T_i for the i-th sorted sample is the conditional expectation within that interval:*

$$T_i = N \int_{z_{i-1}}^{z_i} x\phi(x)\, dx \tag{4.1}$$

where $\phi(x)$ is the standard normal probability density function. Evaluating this integral yields the closed-form target:

$$T_i = N\big(\phi(z_{i-1}) - \phi(z_i)\big). \tag{4.2}$$

This discrete formulation computes the true expected Wasserstein cost only up to an additive constant. To demonstrate this, consider the exact squared Wasserstein distance between the sorted empirical projections y_i and the continuous Gaussian target $\Phi^{-1}(p)$ integrated over the uniform intervals $p \in [\frac{i-1}{N}, \frac{i}{N}]$:

$$W_2^2(P_N, \mathcal{N}(0,1)) = \sum_{i=1}^{N} \int_{\frac{i-1}{N}}^{\frac{i}{N}} \left(y_i - \Phi^{-1}(p)\right)^2 dp. \tag{4.3}$$

Expanding the squared binomial yields the empirical variance $\frac{1}{N}y_i^2$, the target Gaussian variance $\int \Phi^{-1}(p)^2\, dp$, and the cross-term $-2y_i \int \Phi^{-1}(p)\, dp$. Because T_i is defined strictly as $N \int \Phi^{-1}(p)\, dp$, the discrete cost $\frac{1}{N}\sum(y_i - T_i)^2$ evaluates the same empirical variance and cross-terms as the continuous integral.

The two cost functions differ only in the final target variance term. Because the target Gaussian distribution is static, its variance is completely independent of the optimization matrix $\mathbf{W}$. Consequently, this difference is an absolute constant that vanishes under differentiation ($\nabla_{\mathbf{W}} C = \mathbf{0}$). By evaluating this analytical target once during initialization, the algorithm generates an accurate continuous Wasserstein gradient while maintaining the efficiency of discrete sorting.

4.2.2 Batched Vectorization for Restarts

Unlike Principal Component Analysis, which yields a unique global solution via singular value decomposition, the ICA optimization landscape is non-convex and characterized by local optima. Identifying the independent components requires multiple random initializations or restarts to explore the space.

Because evaluating the W_2^2 objective requires an $\mathcal{O}(N \log N)$ sorting operation for each candidate component, executing these random restarts sequentially is computationally expensive. We utilize a batched vectorization strategy. We aggregate B random restarts into a single high-dimensional tensor $\mathbf{W}_{\text{batch}} \in \mathbb{R}^{B \times d \times d}$.

This formulation allows the projection $\mathbf{Y} = \mathbf{W}_{\text{batch}}\widetilde{\mathbf{X}}$ to be computed simultaneously for all B trajectories. The elements of the batch tensor are sorted in parallel. The pre-computed analytical target vector $\mathbf{T}$ is broadcast-subtracted across the batch to compute the distances and gradients in a single step.

4.2.3 The Riemannian Gradient

Because the input data is whitened, the unmixing matrix $\mathbf{W}$ must remain in the Orthogonal Group $\mathcal{S}$, satisfying $\mathbf{W}\mathbf{W}^\top = \mathbf{I}$. If we compute the standard Euclidean gradient of the squared Wasserstein distance, this vector points into the unconstrained ambient space $\mathbb{R}^{d \times d}$. Updating the matrix directly using this Euclidean gradient violates the strict orthogonality constraint.

To navigate the geometry of $\mathcal{S}$, we first compute the Euclidean gradient $\mathbf{G}$

of the empirical objective function in the ambient space. Let $\mathbf{Y} = \mathbf{W}\widetilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times B}$ be the projected mini-batch. For each projection component (row) i, there exists an optimal permutation π_i that sorts the empirical observations such that $y_{i,\pi_i(1)} \leq y_{i,\pi_i(2)} \leq \cdots \leq y_{i,\pi_i(B)}$. The total squared Wasserstein objective across all d dimensions is evaluated as:

$$W_2^2(\mathbf{W}) = \frac{1}{B} \sum_{i=1}^{d} \sum_{j=1}^{B} \big(y_{i,\pi_i(j)} - T_j\big)^2. \tag{4.4}$$

To compute the Euclidean gradient $\mathbf{G} = \nabla_{\mathbf{W}} W_2^2$, we differentiate this objective with respect to $\mathbf{W}$. Utilizing the envelope theorem for optimal transport (Santambrogio, 2015, Section 7.2), the variation of the optimal sorting map π_i evaluates to zero. This allows us to differentiate the squared cost while holding the permutations fixed. Taking the partial derivative with respect to a single row vector $\mathbf{w}_i$ yields:

$$\frac{\partial W_2^2}{\partial \mathbf{w}_i} = \frac{2}{B} \sum_{j=1}^{B} \big(y_{i,\pi_i(j)} - T_j\big) \widetilde{\mathbf{x}}_{\pi_i(j)}. \tag{4.5}$$

This expression evaluates the difference between the projection and the target, mapped back to the original data index. Let $\mathbf{T}^{\text{inv}} \in \mathbb{R}^{d \times B}$ be the target matrix where the analytical targets T_j have been inversely permuted for each row to match the original, unsorted data ordering. The full Euclidean gradient expands into the closed-form matrix equation:

$$\mathbf{G} = \nabla_{\mathbf{W}} W_2^2 = \frac{2}{B} \big(\mathbf{Y} - \mathbf{T}^{\text{inv}}\big) \widetilde{\mathbf{X}}_{\text{batch}}^{\top}. \tag{4.6}$$

To ensure the update step respects the orthogonality constraints, we project this Euclidean gradient $\mathbf{G}$ onto the tangent space of the manifold at the current evaluation point $\mathbf{W}$ (Figure 4.1).

Definition 4.2 Riemannian Gradient on the Orthogonal Group. *Given the Euclidean gradient* $\mathbf{G}$ *and the current unmixing matrix* $\mathbf{W}$*, the Riemannian gradient* $\nabla_{\mathcal{S}} W_2^2$ *resulting from the projection onto the tangent space* $T_{\mathbf{W}}\mathcal{S}$ *is given by (Absil et al., 2008, Chapter 3):*

$$\nabla_{\mathcal{S}} W_2^2 = \mathbf{G} - \frac{1}{2} \big(\mathbf{G}\mathbf{W}^{\top} + \mathbf{W}\mathbf{G}^{\top}\big) \mathbf{W}. \tag{4.7}$$

The term $(\mathbf{G}\mathbf{W}^{\top} + \mathbf{W}\mathbf{G}^{\top})$ mathematically isolates the symmetric violation of orthogonality induced by the Euclidean gradient. Subtracting this term from $\mathbf{G}$ ensures the update step strictly aligns with the intrinsic tangent plane of the

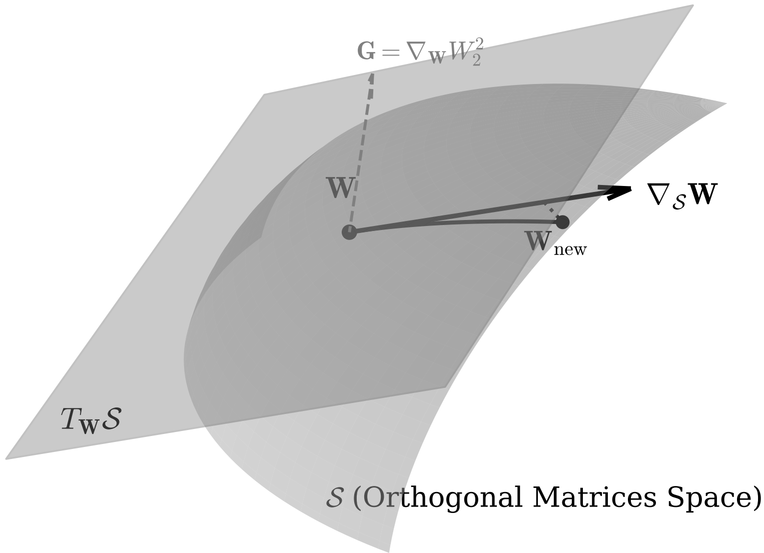

Figure 4.1: *Geometric sketch of optimization on the Orthogonal Group $\mathcal{S}$. The Euclidean gradient $\mathbf{G}$ is projected onto the tangent space $T_{\mathbf{W}}\mathcal{S}$ to yield the Riemannian gradient $\nabla_{\mathcal{S}} W_2^2$ (labeled as the $\nabla_{\mathcal{S}}\mathbf{W}$ update direction in the diagram). Following a step along the tangent plane, a retraction maps the estimate back to the orthogonal surface at $\mathbf{W}_{new}$.*

manifold, preserving the rotational structure of the matrix prior to retraction.

4.2.4 Retraction via Symmetric Decorrelation

Although the Riemannian gradient ensures the update step is tangential to the Orthogonal Group, moving along this tangent plane for a finite step size η causes the new matrix $\mathbf{W}_{\text{step}} = \mathbf{W} - \eta\nabla_{\mathcal{S}}\mathbf{W}$ to depart from the curved surface. To restore exact orthogonality, the matrix must be mapped back onto the manifold via a retraction (Absil et al., 2008, Chapter 4).

Absil, Mahony, and Sepulchre (2008, Chapter 3 and 4) demonstrate that projecting the gradient onto the tangent plane prior to retraction is a numerical requirement. The normal component of the Euclidean gradient $\mathbf{G}$ points directly into the ambient space. Because this vector is orthogonal to the surface, it provides no mathematical progress along the manifold. Furthermore, taking a direct step using $\mathbf{G}$ overshoots into the ambient space, degrading the local approximation bounds required to guarantee the convergence of the retraction mapping (Absil et al., 2008, Chapter 4).

Definition 4.3 Symmetric Decorrelation Retraction. *Let $\mathbf{W}_{step}$ be the unmixing matrix after a tangent space update. To map the matrix back to the orthogonal surface, we first compute its overlap covariance matrix:*

$$\mathbf{C} = \mathbf{W}_{\text{step}}\mathbf{W}_{\text{step}}^{\top}. \tag{4.8}$$

If $\mathbf{W}_{step}$ were strictly orthogonal, $\mathbf{C}$ would equal the identity matrix $\mathbf{I}$. The retraction mapping corrects this deviation by applying the inverse square root of the covariance:

$$\mathbf{W}_{\text{new}} = \mathbf{C}^{-1/2}\mathbf{W}_{\text{step}}. \tag{4.9}$$

We can mathematically verify that this operation successfully retracts the matrix to the Orthogonal Group by evaluating the covariance of the updated matrix:

$$\mathbf{W}_{\text{new}}\mathbf{W}_{\text{new}}^{\top} = (\mathbf{C}^{-1/2}\mathbf{W}_{\text{step}})(\mathbf{W}_{\text{step}}^{\top}\mathbf{C}^{-1/2}) = \mathbf{C}^{-1/2}\mathbf{C}\mathbf{C}^{-1/2} = \mathbf{I}. \tag{4.10}$$

This retraction computes the orthogonal deviation ($\mathbf{C}$) and applies a uniform geometric correction ($\mathbf{C}^{-1/2}$). Because this correction is symmetric, it ensures that no single independent component is prioritized or artificially rotated during the retraction process, maintaining the stability of the batched optimization.

4.3 Total Complexity and Statistical Efficiency

We formalize the total computational cost of the OT-ICA framework. For a dataset of dimension d with N samples, K random restarts, and T optimization iterations, the per-iteration complexity is:

$$\mathcal{O}(K \cdot (d \cdot N \log N + d^3)) \tag{4.11}$$

The $N \log N$ term represents the sorting of projections, while the d^3 term represents the cost of the symmetric decorrelation used for retraction. For typical ICA applications where the number of samples exceeds the dimensionality ($N \gg d$), the d^3 cost is computationally negligible.

Beyond time complexity, we must account for statistical sample complexity. While the 1D projection maintains a convergence rate of $\mathcal{O}(N^{-1/2})$, the global identification of the mixing matrix is implicitly subject to the curse of dimensionality.

Theoretical results in transport theory establish that the empirical Wasserstein distance in d dimensions converges at slower rates as dimension increases (Fournier & Guillin, 2015). As d expands, the number of samples N required to accurately resolve the maxima of non-Gaussianity on the manifold surface grows. An empirical evaluation of this statistical ceiling is reported in Chapter 5.

4.4 Optimization in Discrete Spaces

While the OT-ICA framework demonstrates convergence on continuous distributions, data often contains discrete signals, such as binary states or count-based events. Applying continuous optimal transport directly to discrete mixtures introduces geometric properties that disrupt gradient calculation.

4.4.1 Continuous Smoothing via Gaussian Dithering

We implement a Gaussian Dithering step to restore smooth gradients without resorting to $\mathcal{O}(N^2)$ entropic regularization techniques, such as Sinkhorn distances (Cuturi, 2013).

Adapted from signal processing techniques used to decorrelate quantization error (Schuchman, 1964), we inject a variance of continuous, zero-mean Gaussian noise ($\sigma^2_{\text{dither}} = 0.01$) into the 1D projected components $\mathbf{Y}$ prior to the sorting operation. The sorting operation acts as a strict discrete quantizer. When data clusters densely, ties create highly correlated, deterministic errors in the gradient. By injecting Gaussian dither before sorting, we decorrelate this deterministic quantization error, transforming it into independent noise that first-order gradient solvers can average out. Mathematically, this operation convolves the discrete empirical distribution with a Gaussian kernel (Figure 4.2).

The choice of $\sigma^2_{\text{dither}} = 0.01$ functions as a strictly bounded heuristic governed by the algorithmic pipeline. Because the input data is whitened, its covariance matrix is the identity matrix ($\mathbb{E}[\widetilde{\mathbf{x}}\widetilde{\mathbf{x}}^\top] = \mathbf{I}$). For any 1D projection component $\mathbf{y}_i = \mathbf{w}_i^\top \widetilde{\mathbf{x}}$, the variance is formulated as:

$$\mathrm{Var}(\mathbf{y}_i) = \mathbf{w}_i^\top \mathbb{E}[\widetilde{\mathbf{x}}\widetilde{\mathbf{x}}^\top]\mathbf{w}_i = \mathbf{w}_i^\top \mathbf{I}\mathbf{w}_i = \|\mathbf{w}_i\|^2. \tag{4.12}$$

Furthermore, because the unmixing matrix is restricted to the Orthogonal Group, its row vectors must satisfy the strict orthogonality constraint:

$$\mathbf{W}\mathbf{W}^\top = \mathbf{I} \implies \|\mathbf{w}_i\|^2 = 1. \tag{4.13}$$

Therefore, the variance of any 1D projection within $\mathbf{Y}$ is locked to exactly

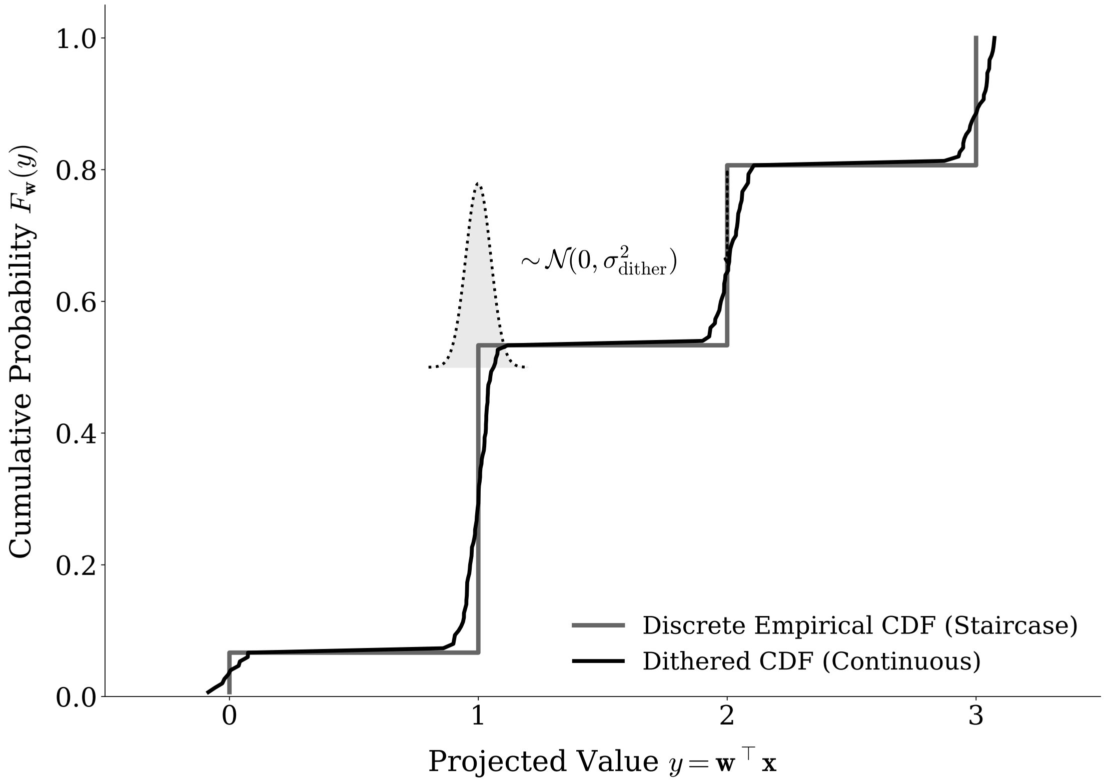

Figure 4.2: *The effect of algorithmic Gaussian Dithering. The non-differentiable staircase of a discrete empirical CDF (gray) is convolved with a narrow Gaussian kernel, yielding a strictly increasing, continuous curve (black).*

1.0. Consequently, a dither variance of 0.01 corresponds to a standard deviation of 0.1, injecting noise scaled to 10% of the projection's standard deviation. This amplitude is sufficient to break deterministic sorting ties at the microscopic level, yet small enough to preserve the macroscopic geometric divergence from Gaussianity measured by the Wasserstein metric.

This deforms the step-wise discrete CDF into a continuous curve. Gaussian dithering restores stable gradients while preserving the $\mathcal{O}(N \log N)$ efficiency of the 1D optimal transport formulation. This acts as an uninformed regularizer applied to the projections $\mathbf{Y}$.

4.4.2 Escaping Local Minima with Stochastic Mini-Batching

Even with a smoothed CDF, the global optimization landscape of a discrete mixture remains non-convex. Evaluating the exact W_2 distance using the full dataset guarantees deterministic convergence to the nearest local maximum. In non-convex environments, this deterministic behavior prevents global convergence.

We utilize stochastic mini-batching. At each optimization step, the algorithm evaluates the Wasserstein gradients using a randomly sampled subset of the data (e.g., $N_{\text{batch}} = 512$). This stochastic subsampling introduces gradient noise. A geometric local minimum that exists for the full dataset presents a different gradient profile for a random subset. This stochastic momentum allows the optimizer to escape spurious local minima.

4.5 OT-ICA Algorithm Overview

The theoretical components and algorithmic enhancements are synthesized into the final OT-ICA procedure. The pseudo-code for extracting independent components using stochastic Riemannian optimization is detailed below (Algorithm 1).

Algorithm 1 The Optimal Transport ICA (OT-ICA) Algorithm

Require: Observed mixture $\mathbf{X} \in \mathbb{R}^{d \times N}$, iterations K, learning rate η, batch size B, dither noise σ_{dither}

Ensure: Estimated unmixing matrix $\mathbf{W}_{\text{final}} \in \mathbb{R}^{d \times d}$

Phase 1: Preprocessing & Target Generation

Center the data: $\mathbf{X} \leftarrow \mathbf{X} - \mathbb{E}[\mathbf{X}]$

Whiten the data: $\widetilde{\mathbf{X}} \leftarrow (\mathbf{X}\mathbf{X}^\top)^{-1/2}\mathbf{X}$ ▷ Yields unit variance, uncorrelated data

Compute analytical target $\mathbf{T} \in \mathbb{R}^B$ via analytical Gaussian CDF integration

Phase 2: Initialization

Initialize $\mathbf{W} \in \mathbb{R}^{d \times d}$ randomly from $\mathcal{N}(0, 1)$

Apply symmetric decorrelation:

$\mathbf{W} \leftarrow (\mathbf{W}\mathbf{W}^\top)^{-1/2}\mathbf{W}$ ▷ Initialize in the Orthogonal Group $\mathcal{S}$

Phase 3: Stochastic Riemannian Optimization

for $k = 1$ to K **do**

Sample stochastic mini-batch $\widetilde{\mathbf{X}}_{\text{batch}} \in \mathbb{R}^{d \times B}$ from $\widetilde{\mathbf{X}}$

Project data onto candidates: $\mathbf{Y} \leftarrow \mathbf{W}\widetilde{\mathbf{X}}_{\text{batch}}$

Apply Gaussian Dithering: $\mathbf{Y}_{\text{dither}} \leftarrow \mathbf{Y} + \mathcal{N}(0, \sigma^2_{\text{dither}})$

Sort each row of $\mathbf{Y}_{\text{dither}}$ ascending to yield empirical quantiles $\mathbf{Y}_{\text{sorted}}$

Compute Wasserstein objective:

$\mathcal{L} = \frac{1}{B}\sum_{i=1}^{d} ||\mathbf{Y}^{(i)}_{\text{sorted}} - \mathbf{T}||_2^2$

Backpropagate to compute Euclidean gradient: $\mathbf{G} = \nabla_{\mathbf{W}}\mathcal{L}$

Project to tangent space (Riemannian Gradient):

$\nabla_{\mathcal{S}}\mathbf{W} = \mathbf{G} - \frac{1}{2}(\mathbf{G}\mathbf{W}^\top + \mathbf{W}\mathbf{G}^\top)\mathbf{W}$

Update matrix along tangent plane:

$\mathbf{W}_{\text{step}} = \mathbf{W} + \eta\nabla_{\mathcal{S}}\mathbf{W}$ ▷ Gradient Ascent

Retract to Orthogonal Group $\mathcal{S}$ via symmetric decorrelation:

$\mathbf{W} \leftarrow (\mathbf{W}_{\text{step}}\mathbf{W}^\top_{\text{step}})^{-1/2}\mathbf{W}_{\text{step}}$

Optional: Apply learning rate decay $\eta \leftarrow \eta \cdot 0.99$

end for

Phase 4: Finalization

$\mathbf{W}_{\text{final}} \leftarrow \mathbf{W}(\mathbf{X}\mathbf{X}^\top)^{-1/2}$ ▷ Map back to original data space

return $\mathbf{W}_{\text{final}}$

4.6 Limitations of Fixed-Point Algorithms for OT-ICA

A widely used algorithm for ICA, FastICA (Hyvärinen et al., 2001, Chapter 8), achieves efficient convergence by employing a second-order Newton fixed-point step. FastICA utilizes this approach by maximizing globally differentiable proxy contrast functions. A natural question is whether a similar fixed-point iteration can be derived for the OT-ICA objective (W_2^2).

Computing exact second-order Newton updates for the empirical Wasserstein metric is analytically intractable without introducing explicit density approximation mechanisms. Because the efficiency of 1D optimal transport relies on the empirical CDF generated via sorting, formulating the exact Hessian matrix $\mathbf{H} = \nabla^2_{\mathbf{w}} W_2^2$ requires differentiating the transport map boundaries. This calculation strictly requires the continuous Probability Density Function (PDF) evaluated at the empirical quantiles. We provide the formal derivation showing this intractability in Appendix A.2.

Satisfying this mathematical requirement necessitates density approximation methods, such as Kernel Density Estimation (KDE). These methods introduce statistical noise and additional computational overhead, eating into the efficiency of the sorting-based transport formulation.

Consequently, the chosen strategy for the OT-ICA framework in this work utilizes Quasi-Newton methods, specifically the Limited-memory BFGS (L-BFGS) algorithm (Liu & Nocedal, 1989) adapted for the Orthogonal Group (Absil et al., 2008, Chapter 6). L-BFGS approximates the inverse Hessian curvature strictly by observing the history of first-order Wasserstein gradients, achieving convergence while preserving the efficiency of the sorting-based metric.

Chapter 5

Experimental Evaluation and Applications

5.1 Evaluation Metrics and Methodology

5.1.1 The Amari Performance Index

To empirically evaluate the success of an Independent Component Analysis (ICA) algorithm, a quantitative metric is required to measure the distance between the estimated unmixing matrix $\mathbf{W}_{\text{est}}$ and the true mixing matrix $\mathbf{A}$. Because ICA is subject to inherent scaling and permutation ambiguities, we utilize the Amari Performance Index (Amari et al., 1996).

Definition 5.1 Amari Performance Index. *Let $\mathbf{P} = \mathbf{W}_{est}\mathbf{A}$ be the global transfer matrix. For a $d \times d$ matrix $\mathbf{P}$ with elements p_{ij}, the Amari error measures the structural divergence from a generalized permutation matrix:*

$$E(\mathbf{P}) = \frac{1}{2d} \sum_{i=1}^{d} \left(\sum_{j=1}^{d} \frac{|p_{ij}|}{\max_k |p_{ik}|} - 1 \right) + \frac{1}{2d} \sum_{j=1}^{d} \left(\sum_{i=1}^{d} \frac{|p_{ij}|}{\max_k |p_{kj}|} - 1 \right). \tag{5.1}$$

This formulation normalizes each row and column by their maximum absolute values. If $\mathbf{P}$ is a generalized permutation matrix, indicating successful source separation, the index evaluates to zero. The experimental evaluations employ specific heuristic thresholds to interpret the Amari error (Table 5.1).

5.1.2 Computational Resource Regimes

Because OT-ICA relies on first-order gradient descent over the non-convex Stiefel manifold, its capacity to find the global optimum depends on the computational

Amari Error (E)	Interpretation / Quality of Separation
$E < 0.1$	Excellent to near-perfect source separation.
$0.1 \leq E < 0.3$	Good separation; the structural shape of the underlying sources is highly recoverable.
$0.3 \leq E < 0.5$	Moderate separation; some structural details may be lost.
$E \geq 0.5$	Severe degradation; multiple sources remain significantly mixed.
$E \geq 1.0$	The estimated matrix is effectively a random orthogonal projection with no meaningful relation to the true sources.

Table 5.1: *Heuristic thresholds for interpreting the Amari Performance Index.*

resources allocated to search the space, particularly as dimensionality increases. Conversely, FastICA utilizes a Newton-based fixed-point iteration that converges rapidly under standard conditions. FastICA requires greater computational resources in certain edge cases where its structural assumptions are violated, causing the solver to oscillate.

To accommodate OT-ICA's scaling requirements, and to evaluate whether FastICA's non-convergence in certain cases is due to mathematical properties rather than standard optimization timeouts, high-dimensional tests in this chapter are evaluated under two computational regimes. The hyperparameter configurations establish specific boundaries for these regimes (Table 5.2).

Regime & Objective	OT-ICA Configuration	FastICA Configuration
Low Compute Baseline *Represents standard, efficient execution.*	**Restarts:** $\min(4 * \#dimensions, 150)$ **Deflation Phase:** 150 iterations **Symmetric Phase:** 200 iterations	**Max Iterations:** 1,000 *Standard limit for fixed-point convergence.*
High Compute Regime *Evaluates structural convergence independent of timeouts.*	**Restarts:** $\min(15 * \#dimensions, 600)$ **Deflation Phase:** 300 iterations **Symmetric Phase:** 500 iterations	**Max Iterations:** 5,000 *Extended to ensure non-convergence is structural.*

Table 5.2: *Summary of the computational regimes used to evaluate algorithmic scaling. These iteration limits and restart thresholds separate standard optimization timeouts from mathematical convergence issues.*

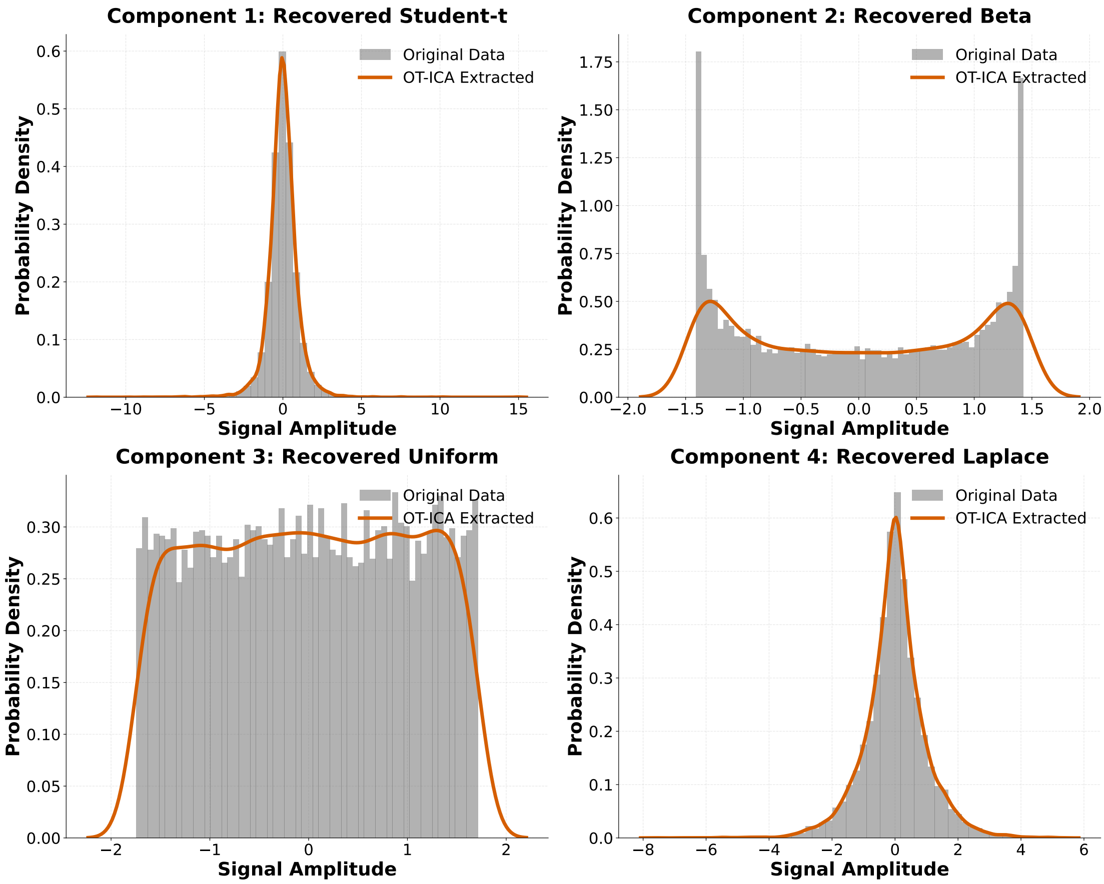

Figure 5.1: *Empirical probability distributions of the 4D baseline validation. The original generated sources are represented by gray histograms, overlaid with the sources extracted by the OT-ICA algorithm (orange curves). The framework recovers continuous non-Gaussian structures, navigating both super-Gaussian peaks and sub-Gaussian bimodality.*

5.2 OT-ICA Methodology Validation

We first verify that OT-ICA successfully recovers a mixture of non-Gaussian distributions in a standard, low-dimensional environment. We generated $N = 10,000$ samples from four continuous independent sources: Laplace (super-Gaussian), Uniform (sub-Gaussian), Student-t (continuous heavy-tailed), and a Beta distribution parameterized at $(0.5, 0.5)$ (bimodal arcsine distribution). These sources were linearly combined using a 4×4 mixing matrix $\mathbf{A}$.

The OT-ICA algorithm successfully extracted the four structural signals from the mixture (Figure 5.1). To quantify this separation numerically, we evaluate the true mixing matrix $\mathbf{A}$, the estimated unmixing matrix $\mathbf{W}_{\text{est}}$, and the global transfer matrix $\mathbf{P} = \mathbf{W}_{\text{est}}\mathbf{A}$. Because ICA is subject to permutation and sign ambiguities, comparing the estimated unmixing matrix to the true mixing matrix directly is challenging. The global transfer matrix $\mathbf{P}$ addresses this; if the algorithm inverted the mixing process, $\mathbf{P}$ approximates a generalized permutation matrix.

The global transfer matrix $\mathbf{P}$ forms a generalized permutation matrix (Table

True Mixing Matrix ($\mathbf{A}$)	Estimated Unmixing Matrix ($\mathbf{W}_{\mathbf{est}}$)
$\begin{bmatrix} 1.058 & 1.520 & -0.249 & 1.012 \\ 0.042 & -0.486 & 0.388 & -0.523 \\ -0.154 & -1.572 & -0.304 & -1.234 \\ -0.006 & 0.503 & 0.053 & 0.928 \end{bmatrix}$	$\begin{bmatrix} 0.154 & -1.943 & 0.608 & -0.456 \\ -0.092 & -0.284 & -0.632 & -1.977 \\ 0.146 & 0.719 & 1.097 & 1.691 \\ 1.021 & 1.204 & 0.828 & 0.677 \end{bmatrix}$
Global Transfer Matrix ($\mathbf{P} = \mathbf{W}_{\mathbf{est}}\mathbf{A}$)	
$\begin{bmatrix} -0.010 & -0.007 & -1.000 & -0.002 \\ 0.000 & -0.003 & 0.000 & -1.000 \\ 0.006 & -1.000 & -0.002 & -0.012 \\ 1.000 & 0.006 & -0.003 & 0.010 \end{bmatrix}$	
Amari Performance Index	
OT-ICA Error: **0.0155**	FastICA Error: **0.0188**

Table 5.3: *The ground truth mixing matrix, estimated unmixing matrix, and resulting global transfer matrix for the OT-ICA baseline validation. The transfer matrix approximates a generalized permutation matrix, resulting in a comparable Amari error to FastICA.*

5.3). This confirms that OT-ICA extracted the independent sources, yielding an Amari error below the 0.1 threshold. For comparison, the FastICA algorithm applied to the same 4D mixture achieved a comparable Amari error.

5.3 Computational & Temporal Scaling

To evaluate computational scaling in higher dimensions, we benchmarked OT-ICA against FastICA using a linear mixture of continuous super-Gaussian Laplacian sources. We evaluated both algorithms across increasing dimensions with a fixed sample size of $N = 10,000$. Because OT-ICA relies on gradient-based optimization over the Stiefel manifold, its capacity to find global optima depends on computational resources. We evaluated both algorithms under the Low-Compute and High-Compute regimes.

FastICA performs well in lower dimensions, maintaining low error across all the compute regimes (Figure 5.2). The Laplacian distribution aligns with FastICA's default *logcosh* contrast function. While OT-ICA scales similarly to FastICA, it requires the High Compute regime to navigate the higher-dimensional optimization landscape. As dimensionality approaches $d = 40$, the statistical resolution of the joint distribution for a sample size of $N = 10,000$ degrades. Both algorithms fail to maintain the $E < 0.3$ threshold, demonstrating a statistical limitation of the fixed sample size.

The computational cost of OT-ICA is significantly higher than that of FastICA (Figure 5.3). FastICA's Newton-based fixed-point iteration allows rapid convergence. In contrast, OT-ICA's first-order gradient descent and parallel restarts cause

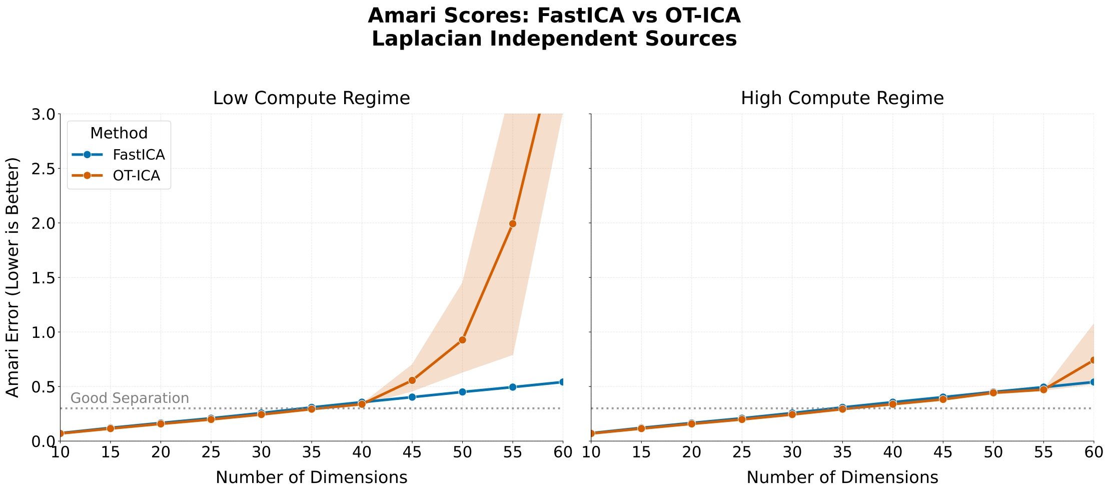

Figure 5.2: *Amari error comparison between FastICA and OT-ICA across dimensions for Laplacian sources. While OT-ICA degrades in the Low Compute regime due to the surface area of the Stiefel manifold, the High Compute regime achieves separation ($E < 0.3$), matching FastICA's capabilities.*

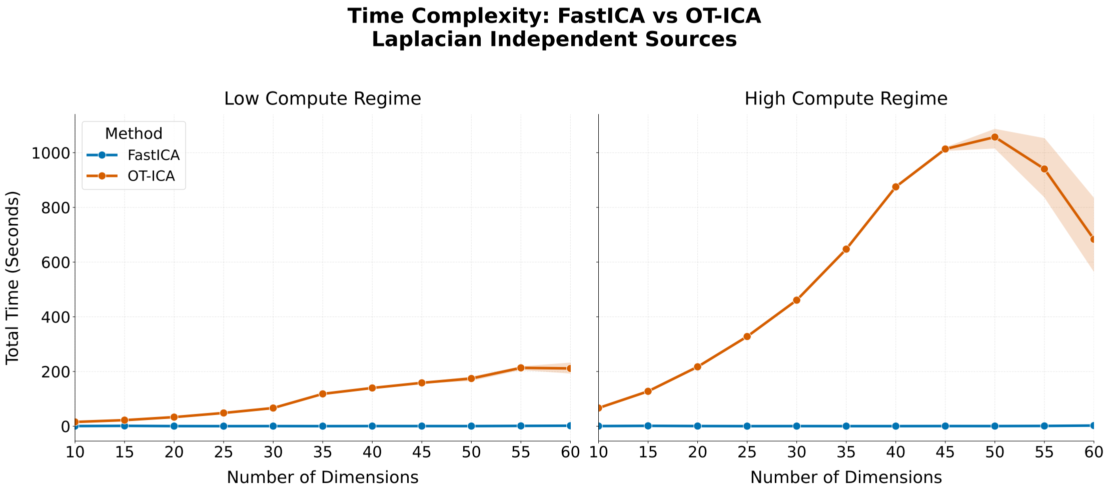

Figure 5.3: *Total execution time (in seconds) for FastICA and OT-ICA across dimensions. OT-ICA exhibits higher computational costs, particularly in the High Compute regime (right) where parallel restarts are required.*

execution times to scale more steeply. This difference in algorithmic efficiency raises the question of whether a gradient-based framework such as OT-ICA offers advantages over FastICA in specific settings.

While FastICA efficiently separates homogeneous mixtures of continuous distributions, empirical data is often heterogeneous. The following sections explore how specific structural properties of non-Gaussian sources affect FastICA's convergence.

5.4 Algorithmic Limitations of FastICA

We examine two specific mathematical conditions that affect FastICA's convergence on certain non-Gaussian independent components, resulting from its reliance on proxy contrast functions and Newton-based iteration.

5.4.1 The Zero Negentropy Condition

FastICA approximates negentropy $J(x)$ using a non-quadratic contrast function $G(\cdot)$, such as the *logcosh* function $G(x) = \frac{1}{a}\log\cosh(ax)$. The approximation is given by:

$$J(x) \propto \big(\mathbb{E}[G(x)] - \mathbb{E}[G(\nu)]\big)^2 \tag{5.2}$$

where ν is a standard Gaussian random variable. The algorithm seeks projections that maximize this squared difference.

If a latent independent source possesses a specific distribution such that its expected value under the contrast function equals that of a Gaussian ($\mathbb{E}[G(x)] = \mathbb{E}[G(\nu)]$), the approximated negentropy evaluates to zero. In this scenario, the objective function provides no gradient signal, and the algorithm does not extract the source.

5.4.2 Empirical Results: Zero Negentropy

To empirically demonstrate this condition, we constructed a symmetric Trimodal Gaussian Mixture parameterized by peak locations $\{-b, 0, b\}$ and variance σ^2. We solved for parameters such that the expected value of the *logcosh* function matched the theoretical baseline of a standard normal distribution. This distribution maintains unit variance, representing an independent non-Gaussian source that yields an approximated negentropy of zero.

We generated linear mixtures of this Trimodal Gaussian source across dimensions ($d \in [5, 25]$) with $N = 10,000$ samples. We separated the evaluation into the Low-Compute and High-Compute regimes.

The empirical results demonstrate that FastICA yields high Amari errors ($E > 0.5$) for this distribution (Figure 5.4). Providing FastICA with High-Compute resources does not improve performance, confirming that the lack of convergence is due to the properties of the proxy objective function rather than to iteration limits.

OT-ICA achieves source separation, by utilizing the empirical CDF via optimal transport and bypassing parametric approximations. While OT-ICA shows performance degradation in the Low Compute regime for $d > 15$ due to the

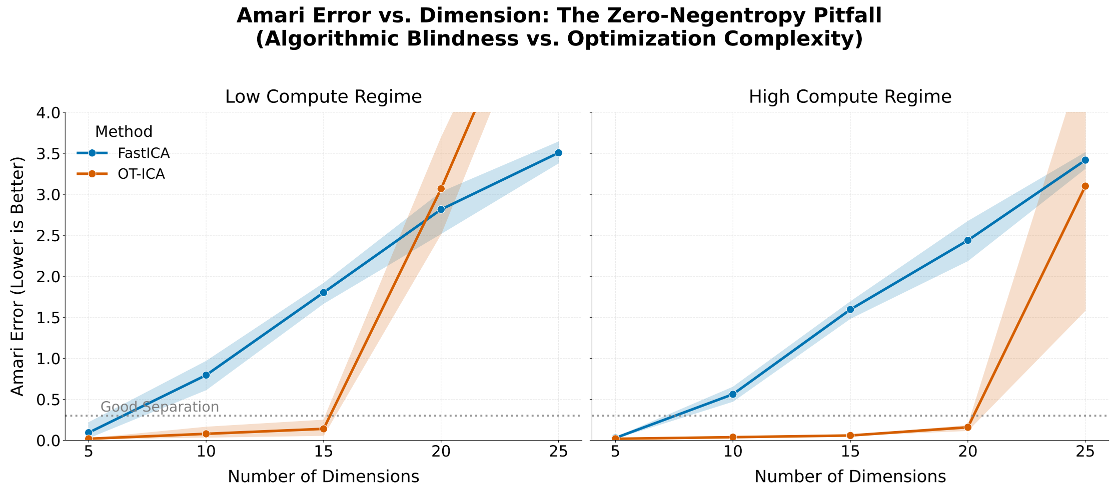

Figure 5.4: *Amari error comparison between FastICA and OT-ICA for the zero-negentropy Trimodal Gaussian distribution. FastICA lacks convergence ($E > 0.5$) across all regimes. OT-ICA unmixes the sources, though it requires the High Compute regime for higher dimensions.*

non-convexity of the Stiefel manifold, the High Compute regime restores accurate unmixing ($E < 0.3$) up to $d = 20$.

5.4.3 The Vanishing Curvature Condition

FastICA utilizes a Newton (second-order) fixed-point iteration step. The mathematical stability of this convergence relies on a condition governing the algorithmic curvature of the optimization landscape, as formalized by Hyvärinen, Karhunen, and Oja (Hyvärinen et al., 2001, Chapter 8) (Appendix A.1).

For the fixed-point iteration to converge to a true independent component s_i, the expectation governing the second-order Taylor expansion of the iteration must not vanish. Defining $g(x) = G'(x)$ and $g'(x) = G''(x)$, where G is the contrast function, convergence requires:

$$\mathbb{E}\big[s_i g(s_i) - g'(s_i)\big] \neq 0 \tag{5.3}$$

For the standard *logcosh* contrast function (with $a = 1$), this equates to $g(x) = \tanh(x)$ and $g'(x) = 1 - \tanh^2(x)$.

If a non-Gaussian source distribution causes this expectation to evaluate to zero, the denominator of the FastICA update rule approaches zero. The fixed-point iteration fails to converge under these conditions.

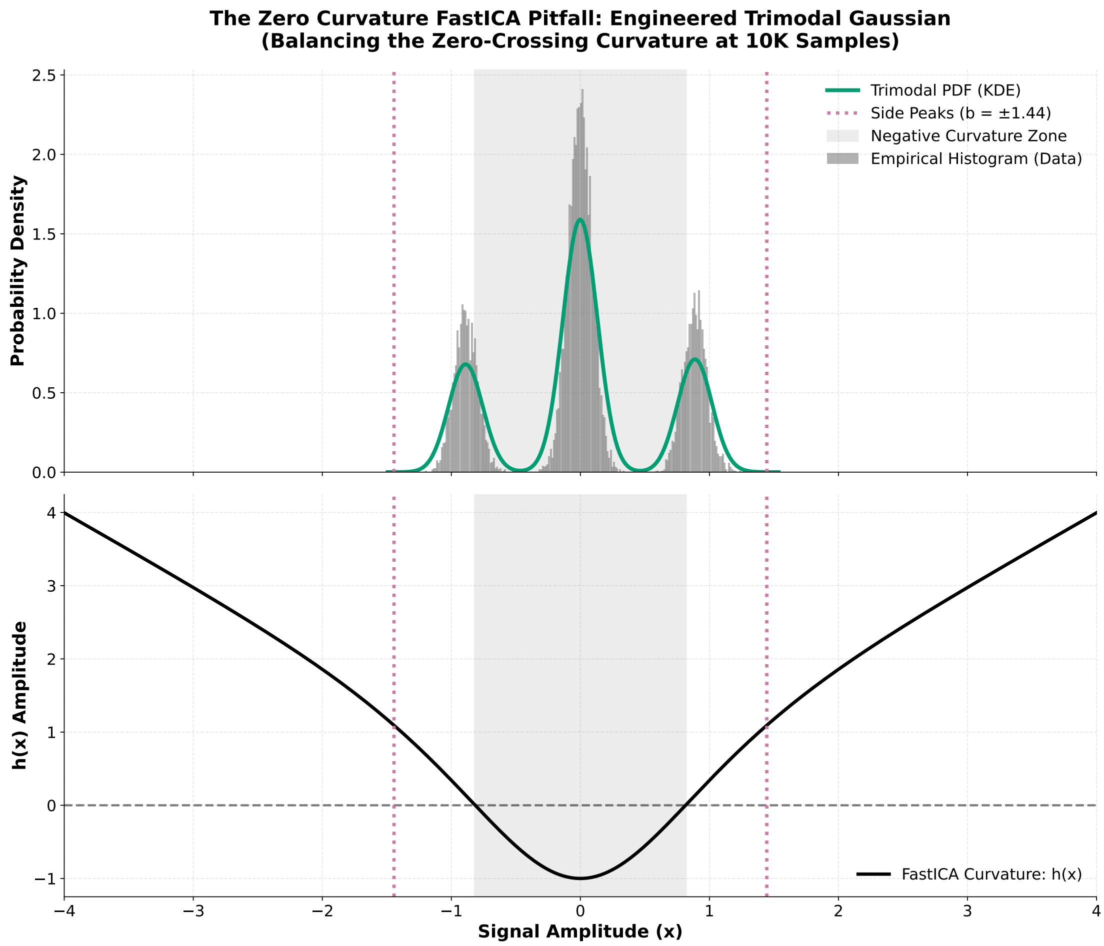

Figure 5.5: *The engineered Trimodal Gaussian Mixture. The central mass generates negative algorithmic curvature, while the side peaks generate positive algorithmic curvature. The parameters b and p are solved such that these regions cancel ($\int h(x)PDF(x)dx = 0$) while maintaining unit variance.*

5.4.4 Empirical Results: Vanishing Curvature

We engineered a continuous source distribution to test this curvature condition. We analyzed the target function:

$$h(x) = x \tanh(x) - 1 + \tanh^2(x). \tag{5.4}$$

We constructed a Trimodal Gaussian Mixture parameterized by symmetrically spaced peak locations $\{-b, 0, b\}$ and variance σ^2. Solving the integral equation $\int h(x)\text{PDF}(x)dx = 0$ subject to the unit variance constraint ($\mathbb{E}[X^2] = 1$) determines parameters that balance the regions of negative and positive curvature.

This distribution is non-Gaussian, but its theoretical algorithmic curvature evaluates to zero (Figure 5.5). We evaluated FastICA and OT-ICA across dimensions ($d \in [5, 25]$) at $N = 10,000$ under both compute regimes.

FastICA returned unmixing matrices with high Amari errors ($E > 0.3$) across

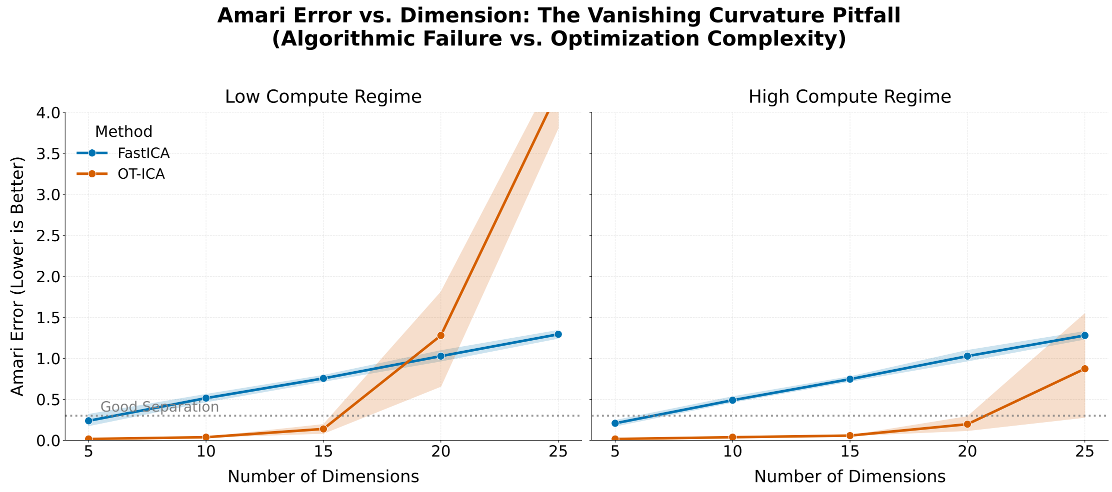

Figure 5.6: *Amari error comparison between FastICA and OT-ICA across dimensions for the engineered Trimodal distribution. FastICA lacks convergence ($E > 1.0$) across all regimes.*

dimensions (Figure 5.6). The High Compute Regime did not improve FastICA's performance, indicating the non-convergence is structural.

OT-ICA unmixed lower dimensions ($d < 20$) in the Low Compute regime and degraded as d increased. In the High Compute regime, OT-ICA maintained separation ($E < 0.3$) through 20 dimensions. This suggests OT-ICA's performance is limited by the expanding non-convex search space rather than local density approximations.

5.5 The Generalized Hybrid Mixture Stress Test

To evaluate OT-ICA and FastICA on heterogeneous signals, we tested a generalized hybrid mixture.

5.5.1 Experimental Setup

We generated linear mixtures with $d = 30$ and $d = 40$ and $N = 10,000$. One source was fixed as a standard Gaussian distribution, $\mathcal{N}(0, 1)$. The remaining $(d-1)$ independent sources were randomly drawn from eight distributions: Laplace, Bernoulli, Uniform, Student-t, Poisson, Binomial, Chi-squared, and Exponential.

FastICA was evaluated with a limit of $10,000$ iterations, while OT-ICA utilized its baseline parameters (Table 5.2).

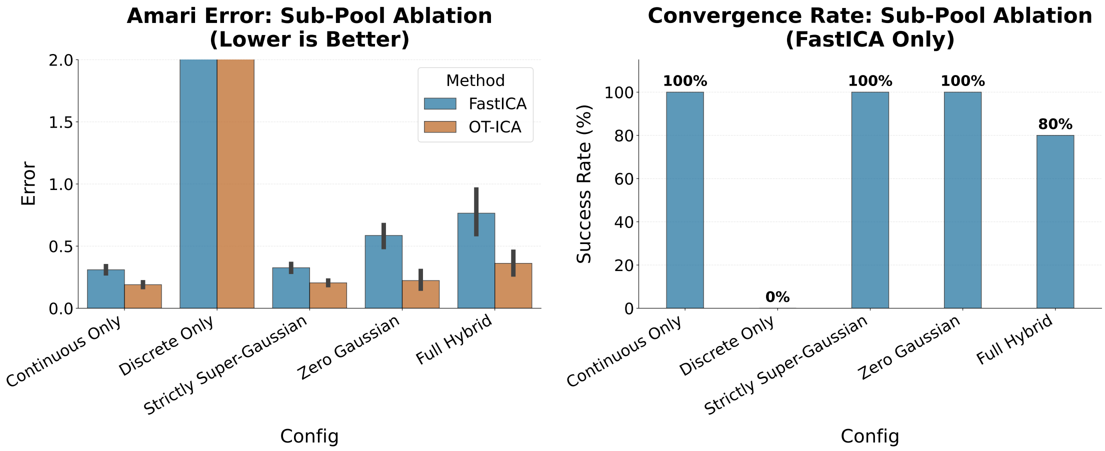

Figure 5.7: *Amari error comparison for the Generalized Hybrid Mixture. OT-ICA maintains stability across the tested dimensions.*

5.5.2 Empirical Results: Hybrid Mixture Stress Test

OT-ICA maintained moderate to good separation ($E < 0.5$) across the dimensions for all but discrete-only mixtures (Figure 5.7).

FastICA exhibited instability in three of the five hybrid mixtures: full hybrid, zero Gaussian, and discrete-only. At $d = 30$, FastICA triggered non-convergence warnings. Providing an extended $10,000$ iteration limit resulted in Amari errors remaining high ($E > 0.5$). This indicates that FastICA's fixed-point update steps may not converge reliably when processing mixtures containing both super-Gaussian and sub-Gaussian signals simultaneously.

5.6 Discrete Only Mixtures: A Unique Challenge

We isolated specific discrete distributions from the discrete-only mixtures to observe their impact on optimization. We generated mixtures containing one continuous Gaussian source and $(d-1)$ sources of a single discrete type (Bernoulli, Poisson, or Binomial).

5.6.1 Empirical Results: Discrete Only Mixtures

FastICA and OT-ICA separated binary Bernoulli mixtures, but both algorithms reported high Amari errors ($E > 1.0$) for multi-step count-based geometries of standard Poisson ($\lambda = 3$) and Binomial ($n = 10$) distributions (Figure 5.8).

5.6.2 Optimization Properties of Discrete Data

To analyze this behavior, we calculated the squared Wasserstein distance (W_2^2) and the Logcosh proxy negentropy to measure the distance between standard normal

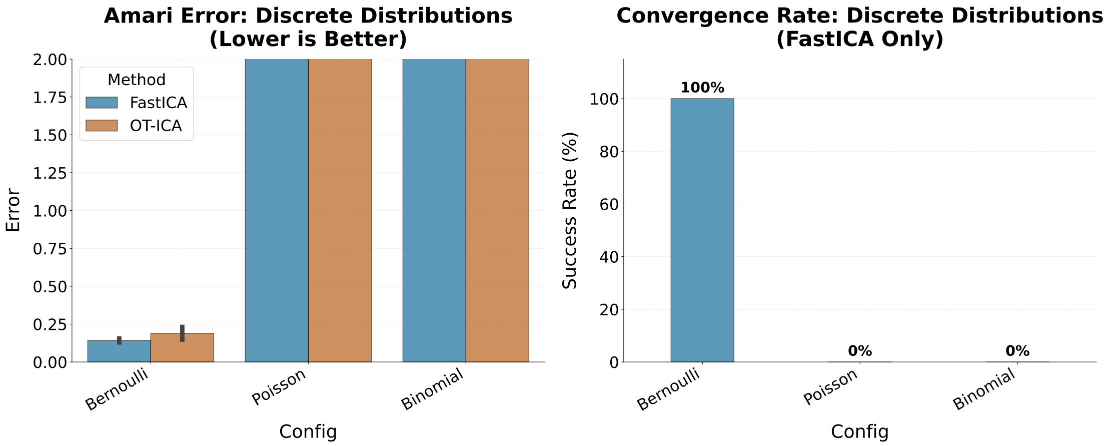

Figure 5.8: *Amari error for mixtures composed entirely of a single discrete distribution type. Both algorithms separate binary (Bernoulli) data but show higher error on count data (Poisson, Binomial).*

noise and discrete distributions.

Distribution	W_2^2 Distance	Logcosh Proxy Negentropy
Laplace (Continuous Baseline)	0.0398	0.001214
Binomial (Standard, $n = 10$)	0.0344	0.000031
Poisson (Standard, $\lambda = 3.0$)	0.0513	0.000000
Binomial (Harsh, $n = 2$)	0.2022	0.000267
Poisson (Harsh, $\lambda = 0.5$)	0.3384	0.000085

Table 5.4: *Comparison of non-Gaussianity metrics across distributions.*

Adjusting the parameters (Poisson $\lambda = 0.5$, Binomial $n = 2$) increases the non-Gaussianity measured by W_2^2 (Table 5.4). The Logcosh proxy yields lower relative values for these discrete structures than the continuous Laplace baseline.

The 1D optimization surfaces were mapped for both metrics (Figure 5.9). For FastICA (dashed blue line), the *logcosh* proxy landscape exhibits low-amplitude gradient changes (scale 10^{-5}). For OT-ICA (solid orange line), the geometric matching reflects the step-function nature of the empirical CDF, creating flat plateaus where the gradient approaches zero. This non-smooth landscape can stall the L-BFGS solver prior to identifying the components.

5.6.3 Empirical Results: Harsh Discrete Environments

We evaluated the algorithms on the skewed parameterizations.

For the skewed Poisson mixture ($\lambda = 0.5$), FastICA did not converge (Figure 5.10). OT-ICA achieved separation ($E \approx 0.15$), as the skewed nature of the Poisson distribution creates a narrower plateau at the minimum of the optimization surface

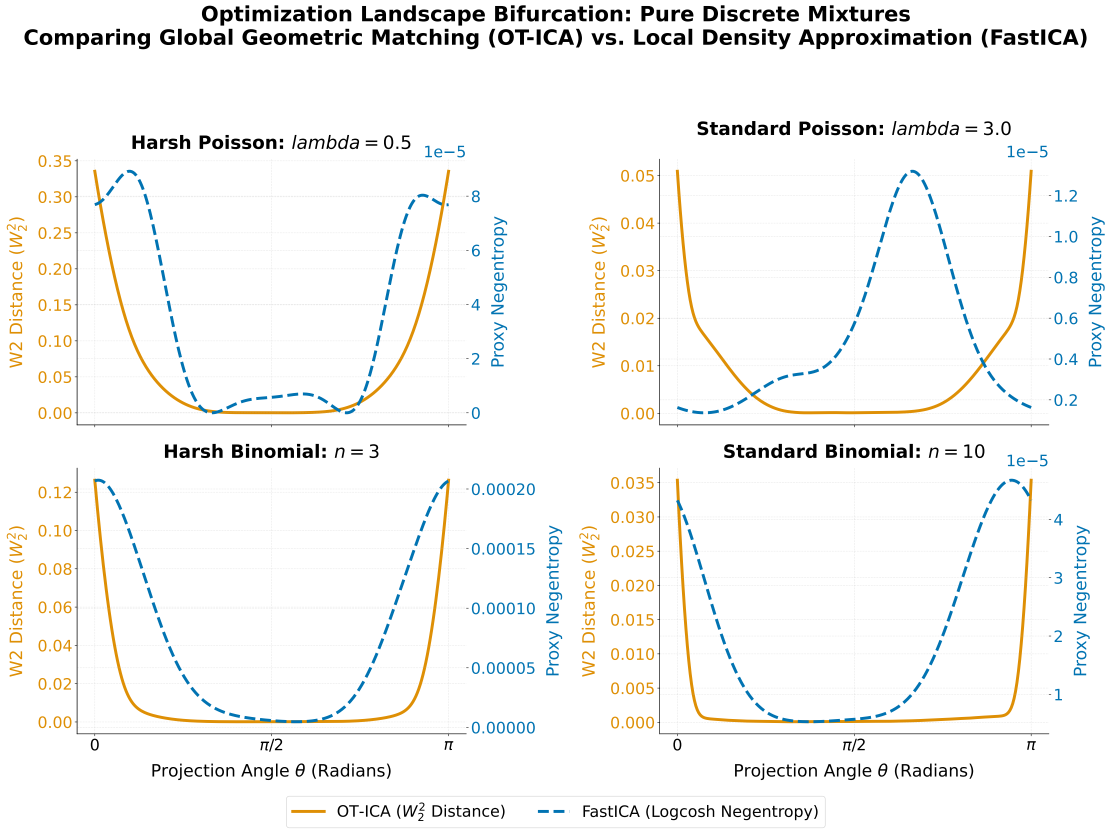

Figure 5.9: *The 1D optimization landscapes for discrete mixtures. The step-function geometry of discrete data affects the gradient properties of both the OT-ICA surface and the FastICA proxy approximation.*

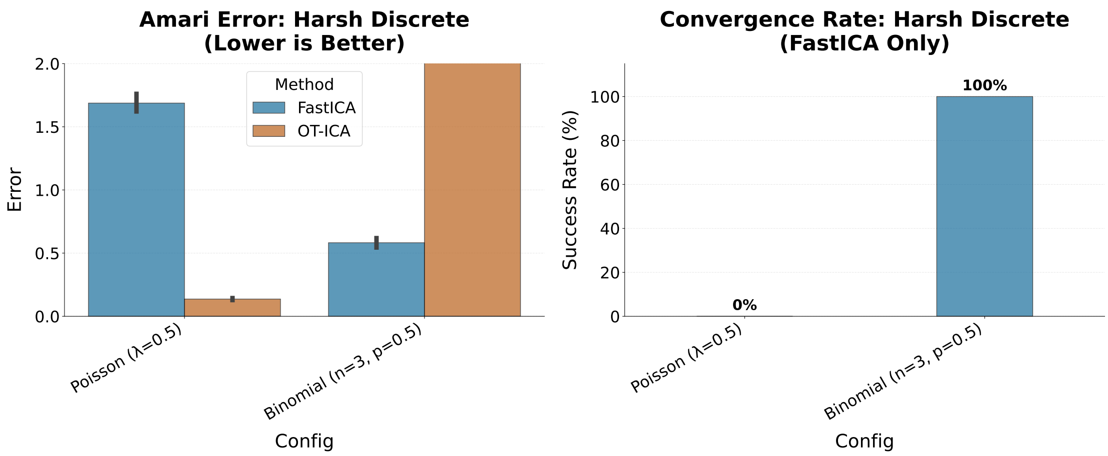

Figure 5.10: *Amari Error for skewed discrete parameterizations. OT-ICA achieves separation on the Poisson distribution but does not converge on the symmetric Binomial.*

(Figure 5.9). For the symmetric Binomial mixture ($n = 3$), OT-ICA got stuck, resulting in an Amari error of $E > 2.0$, due to the wider plateau at the optimum.

5.7 Synthesis: The Case for OT-ICA

The evaluations indicate a distinction between theoretical identifiability and optimization behavior. While statistical non-Gaussianity permits source separation, extracting discrete sources requires navigating a non-smooth optimization surface. As observed in pure discrete environments, step functions create zero-gradient plateaus that stall first-order solvers, despite the objective function correctly identifying the non-Gaussianity.

In heterogeneous environments containing both continuous and discrete variables, the continuous distributions smooth the overall joint Cumulative Distribution Function, allowing the gradient solver to utilize the optimal transport metric across the manifold (Figure 5.7).

FastICA's reliance on local negentropy approximations introduces specific convergence limitations, such as the zero-negentropy condition and vanishing algorithmic curvature, that affect its performance on generalized hybrid mixtures. Because OT-ICA explicitly maps the full empirical geometry via the CDF, it bypasses these specific parametric approximations. Its performance is instead constrained by computational resources and the non-convexity of the Stiefel manifold. For homogeneous mixtures of standard continuous distributions, FastICA provides a computational advantage. In heterogeneous environments where proxy assumptions do not hold, OT-ICA provides a reliable alternative methodology capable of resolving signals that traditional proxy solvers cannot separate.

5.8 Application: EEG Artifact Removal

5.8.1 The Challenge of Volume Conduction

In electroencephalography (EEG) recordings, the human skull acts as a linear volume conductor. Electrical signals generated by brain regions and non-brain muscles (such as the eyes) mix linearly before reaching the scalp electrodes. This satisfies the linear mixing model of ICA: $\mathbf{X} = \mathbf{AS}$.

Eye blinks (ocular artifacts) possess high amplitude and mask underlying brain waves. Because eye blinks are sparse and super-Gaussian, ICA is utilized to isolate and remove them. We apply the OT-ICA framework to a clinical dataset to isolate these artifacts from continuous brain wave recordings.

5.8.2 Empirical Results on Clinical EEG Data

We utilized the `mne` library to process a clinical MEG/EEG dataset. The data was band-pass filtered between 1 Hz and 40 Hz, and we isolated a 10-second window

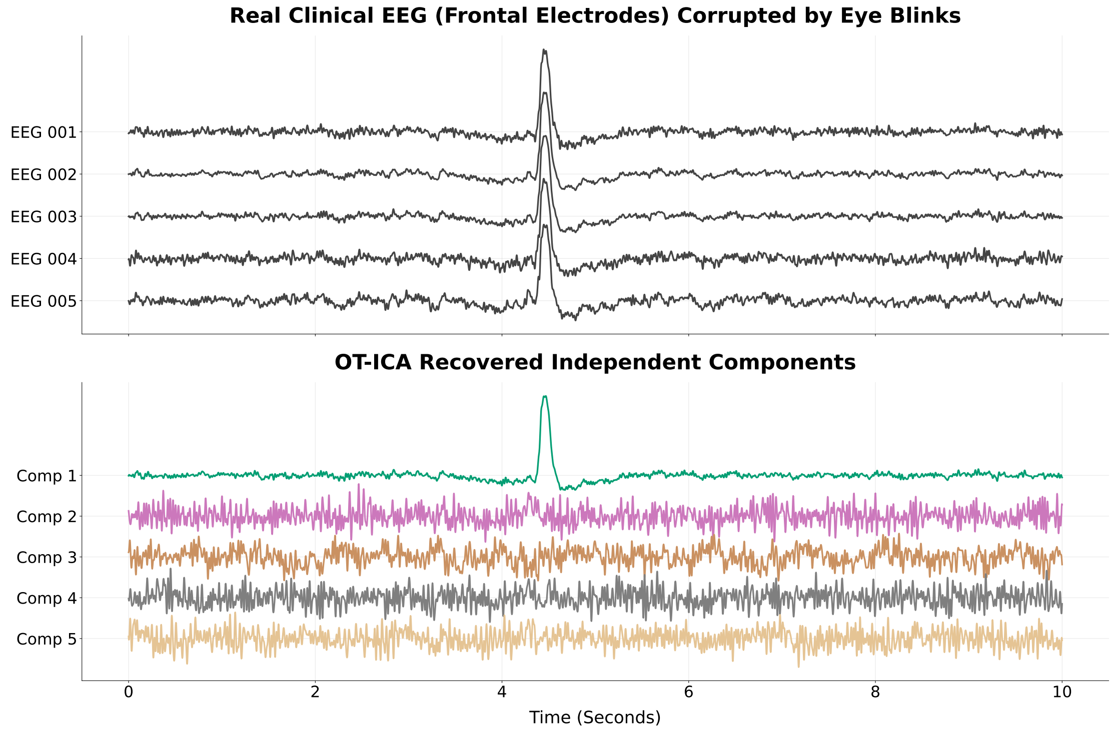

Figure 5.11: *Application of OT-ICA to EEG data.* ***Top (Plot A):*** *Five frontal EEG channels containing eye-blink artifacts.* ***Bottom (Plot B):*** *The independent components recovered by OT-ICA. The ocular artifact is isolated into a single independent component (Comp 1).*

(10.0s to 20.0s) containing eye-blinks across five frontal EEG channels. The data was standardized before applying the OT-ICA algorithm.

The raw frontal electrodes contain synchronized high-amplitude spikes (Figure 5.11, Plot A). The OT-ICA framework separated these signals. The ocular artifact is isolated into a single independent component (Comp 1), leaving the underlying continuous signals in the remaining components (Figure 5.11, Plot B).

5.9 Application: Price Discovery and Information Shares

5.9.1 Non-Gaussianity in Econometric Market Microstructure

We apply OT-ICA to price discovery in financial econometrics. In fragmented financial markets, determining which market contributes most to price discovery involves measuring information shares. Standard vector error correction models (VECMs) identify information shares using the uncorrelation of price innovations. However, identification relying solely on uncorrelation is statistically equivalent up to an orthogonal transformation (Zema & Cordoni, 2025).

By leveraging the non-Gaussianity of latent market shocks, ICA can resolve this orthogonal ambiguity, thereby providing a measurement of market information shares.

5.9.2 Economic Identification Strategy

To address the permutation and sign ambiguities of ICA, we impose two economic identification constraints.

First, to resolve the permutation ambiguity, we enforce a Dominant Diagonal constraint. The structural shock assigned to Market i must account for the maximum variance in that market's price innovations. We implement this using the Hungarian Algorithm for optimal bipartite matching (Kuhn, 1955).

Second, to resolve the sign ambiguity, we enforce a Positive Impact constraint based on positive spillovers. A structural innovation representing 'good news' is assumed to have a positive initial impact on its own market price. Columns of the unmixing matrix violating this logic are sign-flipped.

5.9.3 Empirical Results on Simulated Market Data

We simulated a fragmented market scenario targeting three markets with pre-defined Information Shares of 12%, 24%, and 64%. Market innovations were generated using Student-t distributions to satisfy the non-Gaussianity requirements. We executed 500 Monte Carlo simulations comprising 5,000 observations each.

A representative run demonstrates the correlation between the generative shocks and the empirical shocks estimated by OT-ICA (Figure 5.12).

Applying the Dominant Diagonal and Positive Impact constraints, we mapped the statistical components to their corresponding markets and calculated empirical information shares.

The estimated information shares converge to their theoretical targets across all 500 simulations (Table 5.5).

Market / Source	**True IS**	**Estimated IS (Mean)**	**Standard Deviation**
Market 1	0.1200	0.1212	0.0161
Market 2	0.2400	0.2399	0.0245
Market 3	0.6400	0.6389	0.0260

Table 5.5: *Monte Carlo simulation results comparing true Information Shares (IS) against the estimated IS recovered via OT-ICA over 500 runs.*

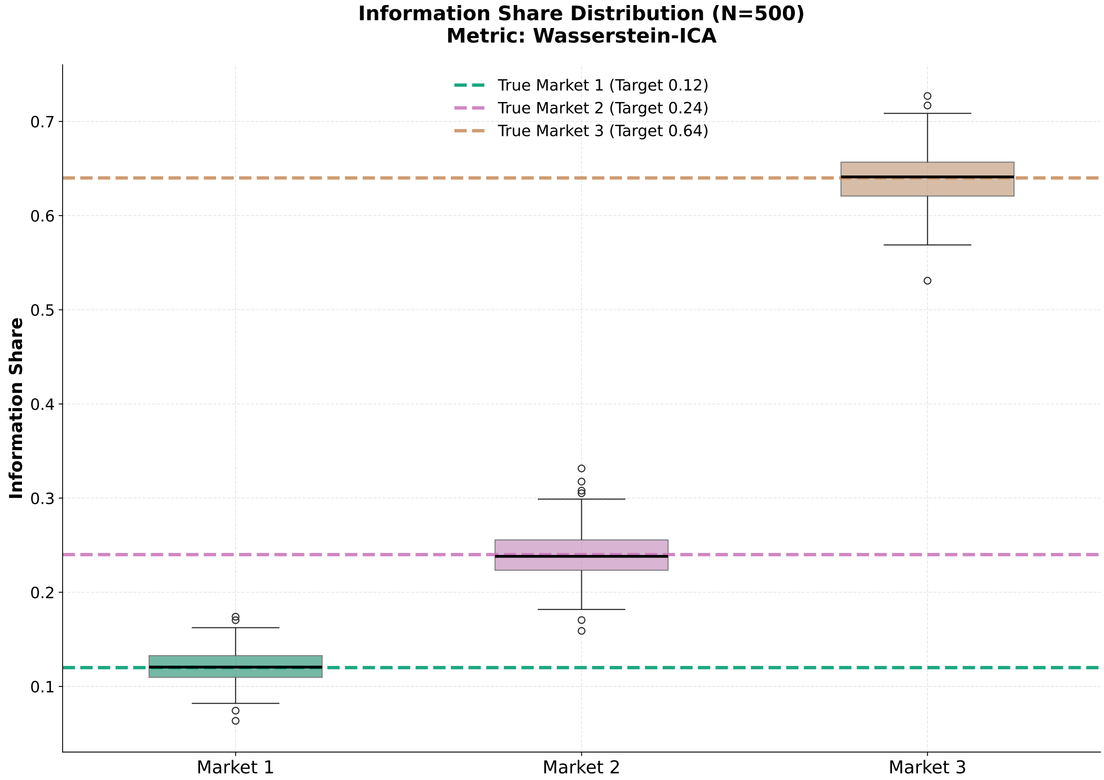

Figure 5.12: *Pairplot demonstrating the recovery of true structural shocks versus estimated shocks from OT-ICA for a single Monte Carlo run. The linear alignment indicates unmixing of the latent market innovations.*

The empirical density of the estimated information shares is centered around the true parameter values across all simulation runs (Figure 5.13).

These empirical results demonstrate the viability of the OT-ICA framework for price discovery applications. Standard VECMs rely solely on decorrelation, leaving structural shocks unidentified. However, by exploiting the non-Gaussian properties of market innovations and applying basic economic constraints, OT-ICA identifies these structural elements. We therefore conclude that OT-ICA provides a reliable framework for measuring price discovery across markets.

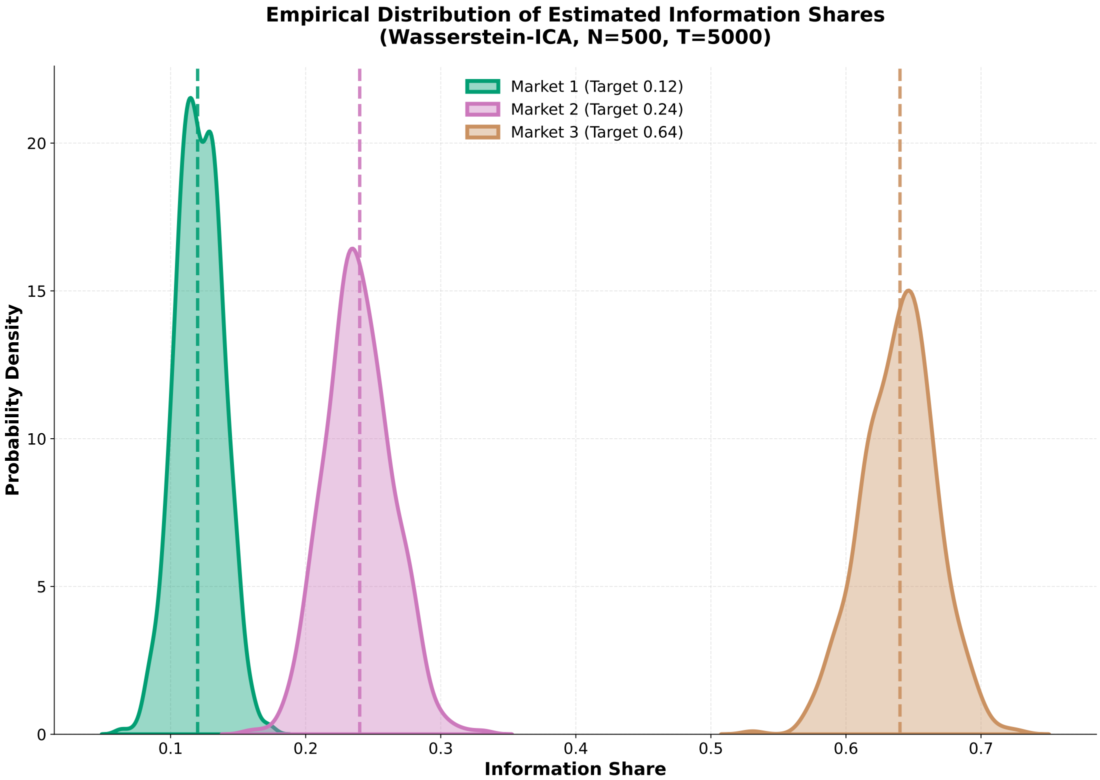

Figure 5.13: *Empirical density distributions of the estimated Information Shares across 500 Monte Carlo simulations. The vertical dashed lines represent the true theoretical values.*

Chapter 6

Conclusion

6.1 Summary of Contributions

This thesis evaluated the Optimal Transport Independent Component Analysis (OT-ICA) framework, transitioning linear source separation from local proxy approximations to global geometric matching. By formulating the search for independent components as minimizing the squared Wasserstein distance (W_2^2) to a standard normal distribution on the Stiefel manifold, we addressed specific mathematical vulnerabilities present in established proxy algorithms.

Our theoretical analysis established that the squared Wasserstein distance bounds mixture non-Gaussianity, providing a direct geometric objective for extracting latent sources. Empirically, we demonstrated that while Newton-based algorithms like FastICA fail to converge in the presence of structural blind spots, such as zero-negentropy topologies and vanishing curvature, OT-ICA maintains stable convergence. By mapping the full empirical cumulative distribution function rather than relying on point-estimates of density, OT-ICA successfully resolved generalized hybrid mixtures that would otherwise cause proxy solvers to oscillate indefinitely.

We introduced specific algorithmic adaptations to make this optimal transport framework computationally viable. These included exact analytical Gaussian targets to eliminate discretization noise, batched vectorization to handle the non-convex search space, and Gaussian dithering to process the discontinuous optimization landscapes of discrete data. Finally, we demonstrated the practical application of this framework, utilizing OT-ICA to isolate sparse artifacts in clinical EEG recordings and to resolve orthogonal ambiguities in econometric price discovery.

6.2 Limitations

The OT-ICA framework is subject to constraints regarding computational efficiency. Because computing the exact Hessian of the Wasserstein distance relies on continuous density estimation, OT-ICA must use first-order Quasi-Newton methods (L-BFGS) with parallel random restarts. This results in execution times that scale exponentially with dimensionality, making it computationally expensive compared to fixed-point iteration solvers.

The framework also encounters a statistical performance ceiling related to the curse of dimensionality. As the dimensionality expands, the non-convex search space of the orthogonal group $O(d)$ becomes increasingly complex. In our empirical evaluations, a fixed sample size of $N = 10,000$ became insufficient to maintain the heuristic threshold for good source separation (Amari error $E < 0.3$) when $d > 40$, demonstrating a statistical resolution limit of the joint distribution.

Finally, while Gaussian dithering smooths step-wise empirical CDFs, highly symmetric and sparse discrete distributions (such as a low parameter Binomial distribution) yield optimization landscapes with flat, zero-gradient plateaus. While OT-ICA navigates hybrid environments where continuous variables smooth the joint distribution, isolating purely discrete, symmetrically clustered sources remains a limitation for first-order gradient solvers.

6.3 Future Work

Future research directions include the integration of entropic regularization (Sinkhorn distances) (Cuturi, 2013; Peyré & Cuturi, 2019) to replace the Gaussian dithering step. Sinkhorn divergence provides a naturally differentiable approximation of the Wasserstein metric that processes discrete distributions without mathematically altering the underlying data.

Improving computational scaling represents another critical objective. Investigating stochastic second-order methods, or utilizing neural-based transport map approximations (such as Normalizing Flows), could reduce the execution time required to search the Stiefel manifold. Leveraging Amortized Optimal Transport to learn the transport map offline would bypass the $\mathcal{O}(N \log N)$ sorting requirement entirely during runtime.

Integrating the OT-ICA framework into the domain of causal discovery offers another promising avenue. Causal structures are frequently modeled using Linear Non-Gaussian Acyclic Models (LinGAM) (Shimizu, Hoyer, Hyvärinen, & Kerminen, 2006). Because LinGAM relies directly on ICA to identify the independent error terms that dictate causal directionality, applying a globally geometrically aware metric like OT-ICA could improve causal estimation in complex, heterogeneous datasets where traditional ICA proxies fail. Furthermore, the emerging field of Causal Representation Learning seeks to infer causally related latent variables. The recently proposed Causal Component Analysis (CauCA) (Liang et al., 2023) bridges ICA and causal learning by modeling the causal dependence among latent components under known graphs. Integrating the robust Wasserstein metric of OT-ICA into the CauCA framework could enhance the estimation of unmixing functions and causal mechanisms, particularly when latent variables exhibit complex, heterogeneous non-Gaussianity.

Finally, extending the geometric properties of optimal transport contrast functions to nonlinear Independent Component Analysis represents a compelling frontier. Unsupervised learning of latent variable models via nonlinear ICA requires strict constraints on the function class to ensure identifiability, as explored by Buchholz, Besserve, and Schölkopf (Buchholz, Besserve, & Schölkopf, 2022). The optimal transport metric, which inherently evaluates global structural deformations, may serve to effectively regularize these constrained function spaces, enabling reliable unmixing of nonlinear representations.

Appendix A

Mathematical Proofs and Derivations

A.1 Derivation of the FastICA Fixed-Point Failure Condition

Proof of Theorem 8.1 (Hyvärinen et al., 2001, Chapter 8) Denote by $H(\mathbf{w})$ the function to be minimized/maximized, $E\{G(\mathbf{w}^T\mathbf{z})\}$. Make the orthogonal change of coordinates $\mathbf{q} = \mathbf{A}^T\mathbf{V}^T\mathbf{w}$. Then we can calculate the gradient as $\frac{\partial H(\mathbf{q})}{\partial \mathbf{q}} = E\{\mathbf{s}g(\mathbf{q}^T\mathbf{s})\}$ and the Hessian as $\frac{\partial^2 H(\mathbf{q})}{\partial \mathbf{q}^2} = E\{\mathbf{s}\mathbf{s}^T g'(\mathbf{q}^T\mathbf{s})\}$. Without loss of generality, it is enough to analyze the stability of the point $\mathbf{q} = \mathbf{e}_1$, where $\mathbf{e}_1 = (1, 0, 0, 0, \dots)$. Evaluating the gradient and the Hessian at point $\mathbf{q} = \mathbf{e}_1$, we get using the independence of the s_i,

$$\frac{\partial H(\mathbf{e}_1)}{\partial \mathbf{q}} = \mathbf{e}_1 E\{s_1 g(s_1)\} \tag{A.1}$$

and

$$\frac{\partial^2 H(\mathbf{e}_1)}{\partial \mathbf{q}^2} = \mathrm{diag}(E\{s_1^2 g'(s_1)\}, E\{g'(s_1)\}, E\{g'(s_1)\}, \dots). \tag{A.2}$$

Making a small perturbation $\epsilon = (\epsilon_1, \epsilon_2, \dots)$, we obtain

$$\begin{aligned} H(\mathbf{e}_1 + \epsilon) &= H(\mathbf{e}_1) + \epsilon^T \frac{\partial H(\mathbf{e}_1)}{\partial \mathbf{q}} + \frac{1}{2}\epsilon^T \frac{\partial^2 H(\mathbf{e}_1)}{\partial \mathbf{q}^2}\epsilon + o(\|\epsilon\|^2) \\ &= H(\mathbf{e}_1) + E\{s_1 g(s_1)\}\epsilon_1 + \frac{1}{2}\Big[E\{s_1^2 g'(s_1)\}\epsilon_1^2 + E\{g'(s_1)\}\sum_{i>1}\epsilon_i^2\Big] + o(\|\epsilon\|^2) \end{aligned} \tag{A.3}$$

Due to the constraint $\|\mathbf{w}\| = 1$ we get $\epsilon_1 = \sqrt{1 - \epsilon_2^2 - \epsilon_3^2 - \dots} - 1$. Due to the fact that $\sqrt{1-\gamma} = 1 - \gamma/2 + o(\gamma)$, the term of order ϵ_1^2 is $o(\|\epsilon\|^2)$, i.e., of higher order, and can be neglected. Using the aforementioned first-order approximation for ϵ_1

we obtain $\epsilon_1 = -\sum_{i>1} \epsilon_i^2/2 + o(\|\epsilon\|^2)$, which finally gives

$$H(\mathbf{e}_1 + \epsilon) = H(\mathbf{e}_1) + \frac{1}{2}\Big[E\{g'(s_1) - s_1 g(s_1)\}\Big] \sum_{i>1} \epsilon_i^2 + o(\|\epsilon\|^2) \tag{A.4}$$

which clearly proves $\mathbf{q} = \mathbf{e}_1$ is an extremum, and of the type implied by the condition of the theorem.

Proof of convergence of FastICA The convergence is proven under the assumptions that first, the data follows the ICA data model and second, that the expectations are evaluated exactly.

Let g be the nonlinearity used in the algorithm. In the case of the kurtosis-based algorithm, this is the cubic function, so we obtain that algorithm as a special case of the following proof for a general g. We must also make the following technical assumption:

$$E\{s_i g(s_i) - g'(s_i)\} \neq 0, \quad \text{for any } i \tag{A.5}$$

which can be considered a generalization of the condition valid when we use kurtosis, that the kurtosis of the independent components must be nonzero. If this is true for a subset of independent components, we can estimate just those independent components.

To begin with, make the change of variable $\mathbf{q} = \mathbf{A}^T\mathbf{V}^T\mathbf{w}$, as earlier, and assume that $\mathbf{q}$ is in the neighborhood of a solution (say, $q_1 \approx 1$ as before). As shown in the proof of Theorem 8.1, the change in q_1 is then of a lower order than the change in the other coordinates, due to the constraint $\|\mathbf{q}\| = 1$. Then we can expand the terms using a Taylor approximation for g and g', first obtaining

$$\begin{aligned} g(\mathbf{q}^T\mathbf{s}) = g(q_1 s_1) + g'(q_1 s_1)\mathbf{q}_{-1}^T\mathbf{s}_{-1} + \frac{1}{2}g''(q_1 s_1)(\mathbf{q}_{-1}^T\mathbf{s}_{-1})^2 \\ + \frac{1}{6}g'''(q_1 s_1)(\mathbf{q}_{-1}^T\mathbf{s}_{-1})^3 + O(\|\mathbf{q}_{-1}\|^4) \end{aligned} \tag{A.6}$$

and then

$$\begin{aligned} g'(\mathbf{q}^T\mathbf{s}) = g'(q_1 s_1) + g''(q_1 s_1)\mathbf{q}_{-1}^T\mathbf{s}_{-1} \\ + \frac{1}{2}g'''(q_1 s_1)(\mathbf{q}_{-1}^T\mathbf{s}_{-1})^2 + O(\|\mathbf{q}_{-1}\|^3) \end{aligned} \tag{A.7}$$

where $\mathbf{q}_{-1}$ and $\mathbf{s}_{-1}$ are the vectors $\mathbf{q}$ and $\mathbf{s}$ without their first components. Denote by $\mathbf{q}^+$ the new value of $\mathbf{q}$ (after one iteration). Thus we obtain, using the independence of the s_i and doing some tedious but straightforward algebraic manipulations,

$$q_1^+ = E\{s_1 g(q_1 s_1) - g'(q_1 s_1)\} + O(\|\mathbf{q}_{-1}\|^2) \tag{A.8}$$

$$q_i^+ = \frac{1}{2}E\{s_i^3\}E\{g''(s_1)\}q_i^2 + \frac{1}{6}\text{kurt}(s_i)E\{g'''(s_1)\}q_i^3 + O(\|\mathbf{q}_{-1}\|^4), \quad \text{for } i > 1 \tag{A.9}$$

We obtain also

$$\mathbf{q}^* = \mathbf{q}^+/\|\mathbf{q}^+\| \tag{A.10}$$

This shows clearly that under assumption (A.5), the algorithm converges (locally) to such a vector $\mathbf{q}$ that $q_1 = \pm 1$ and $q_i = 0$ for $i > 1$. This means that $\mathbf{w} = ((\mathbf{VA})^T)^{-1}\mathbf{q}$ converges, up to the sign, to one of the rows of the inverse of the mixing matrix $\mathbf{VA}$, which implies that $\mathbf{w}^T\mathbf{z}$ converges to one of the s_i. Moreover, if $E\{g''(s_1)\} = 0$, i.e., if the s_i has a symmetric distribution, as is usually the case, the convergence is cubic. In other cases, the convergence is quadratic. If kurtosis is used, however, we always have $E\{g''(s_1)\} = 0$ and thus cubic convergence. In addition, if $G(y) = y^4$, the local approximations are exact, and the convergence is global.

A.2 Intractability of Exact Newton Updates for Empirical OT-ICA

Theorem A.1 Intractability of Exact Newton Updates. *For an empirical distribution mapped to a continuous target via sorting, computing an exact, closed-form Hessian $\mathbf{H} = \nabla^2_{\mathbf{w}} W_2^2$ is analytically intractable without introducing external density estimation mechanisms. The Map Derivative $\nabla_{\mathbf{w}} T_{\mathbf{w}}$ inherently relies upon the rate of change of cumulative probability mass, which strictly requires the continuous Probability Density Function (PDF) evaluated at the quantile boundaries.*

Proof. To implement a Newton step, the exact Hessian matrix of the squared Wasserstein distance, $\mathbf{H} = \nabla^2_{\mathbf{w}} W_2^2$, is required. We first compute the first derivative of the objective with respect to the weight vector $\mathbf{w}$. Utilizing the envelope theorem for optimal transport (Santambrogio, 2015, Section 7.2), the variation of the optimal map evaluates to zero within the cost function, allowing us to differentiate the squared cost directly:

$$\begin{aligned}
\nabla_{\mathbf{w}}\left(\frac{1}{2}W_2^2\right) &= \nabla_{\mathbf{w}}\left(\frac{1}{2}\mathbb{E}\Big[\big(\mathbf{w}^\top\mathbf{X} - T_{\mathbf{w}}(\mathbf{w}^\top\mathbf{X})\big)^2\Big]\right) \\
&= \mathbb{E}\Big[\big(\mathbf{w}^\top\mathbf{X} - T_{\mathbf{w}}(\mathbf{w}^\top\mathbf{X})\big)\nabla_{\mathbf{w}}(\mathbf{w}^\top\mathbf{X})\Big] \\
&= \mathbb{E}\Big[\mathbf{X}\big(\mathbf{w}^\top\mathbf{X} - T_{\mathbf{w}}(\mathbf{w}^\top\mathbf{X})\big)\Big].
\end{aligned} \tag{A.11}$$

To compute the Hessian, we differentiate the optimal transport map $T_{\mathbf{w}}(\mathbf{w}^\top\mathbf{X})$

with respect to $\mathbf{w}$. Applying the total derivative yields two terms:

$$\nabla_{\mathbf{w}}[T_{\mathbf{w}}(\mathbf{w}^\top \mathbf{X})] = \underbrace{T'_{\mathbf{w}}(\mathbf{w}^\top \mathbf{X})\mathbf{X}}_{\text{Argument Derivative}} + \underbrace{(\nabla_{\mathbf{w}} T_{\mathbf{w}})(\mathbf{w}^\top \mathbf{X})}_{\text{Map Derivative}}. \tag{A.12}$$

By Caffarelli's regularity theory (Caffarelli, 1992), the transport map $T_{\mathbf{w}}$ is continuously differentiable, guaranteeing the existence of the Argument Derivative. The limitation lies within the Map Derivative.

In the 1D optimal transport setting, the transport map to a standard Gaussian relies on the empirical cumulative distribution function, $F_{\mathbf{w}}$, of the projected data $y = \mathbf{w}^\top \mathbf{X}$. Expanding the Map Derivative yields:

$$(\nabla_{\mathbf{w}} T_{\mathbf{w}})(y) = \nabla_{\mathbf{w}} \left[\Phi^{-1}(F_{\mathbf{w}}(y))\right] = \frac{1}{\phi(\Phi^{-1}(F_{\mathbf{w}}(y)))} \nabla_{\mathbf{w}} F_{\mathbf{w}}(y). \tag{A.13}$$

The term $\nabla_{\mathbf{w}} F_{\mathbf{w}}(y)$ represents the rate of change of the cumulative probability mass as the projection plane $\mathbf{w}$ is rotated. The derivative of a cumulative distribution function's boundary strictly requires the underlying probability density function (PDF) evaluated at that boundary, rendering the closed-form calculation intractable without explicit density estimation. ■

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